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(54) **OPTICAL IMAGE LENS ASSEMBLY, IMAGE CAPTURING DEVICE AND ELECTRONIC DEVICE**

(56) **References Cited**

U.S. PATENT DOCUMENTS

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|              |    |         |             |
|--------------|----|---------|-------------|
| 8,724,237    | B2 | 5/2014  | Hsu et al.  |
| 9,046,672    | B2 | 6/2015  | You         |
| 9,274,308    | B2 | 3/2016  | Nabeta      |
| 9,274,315    | B2 | 3/2016  | Lee         |
| 9,482,847    | B1 | 11/2016 | Liu et al.  |
| 2015/0062405 | A1 | 3/2015  | Chen et al. |
| 2015/0062407 | A1 | 3/2015  | Chen et al. |
| 2015/0109684 | A1 | 4/2015  | Son         |
| 2015/0338613 | A1 | 11/2015 | Tang et al. |
| 2015/0338614 | A1 | 11/2015 | Tang et al. |
| 2015/0362702 | A1 | 12/2015 | Tang et al. |
| 2016/0018629 | A1 | 1/2016  | Tang et al. |
| 2016/0097916 | A1 | 4/2016  | Tang et al. |
| 2016/0124184 | A1 | 5/2016  | Tang et al. |

(Continued)

FOREIGN PATENT DOCUMENTS

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See application file for complete search history.

CN 105044880 A 11/2015  
TW 201413320 A 4/2014

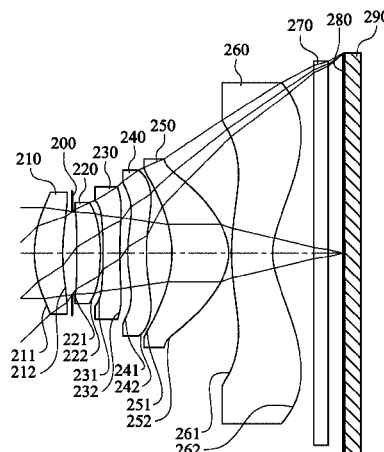
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(57) **ABSTRACT**

An optical image lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element and a sixth lens element. The first lens element with positive refractive power has an object-side surface being convex. The second lens element with positive refractive power has an image-side surface being convex. The fourth lens element has an image-side surface being concave. The fifth lens element with positive refractive power has an object-side surface being concave and an image-side surface being convex. The sixth lens element with negative refractive power has an image-side surface being concave in a paraxial region thereof and including at least one convex shape in an off-axial region thereof. The two surfaces of each of the fifth lens element and the sixth lens element are aspheric.

**16 Claims, 19 Drawing Sheets**

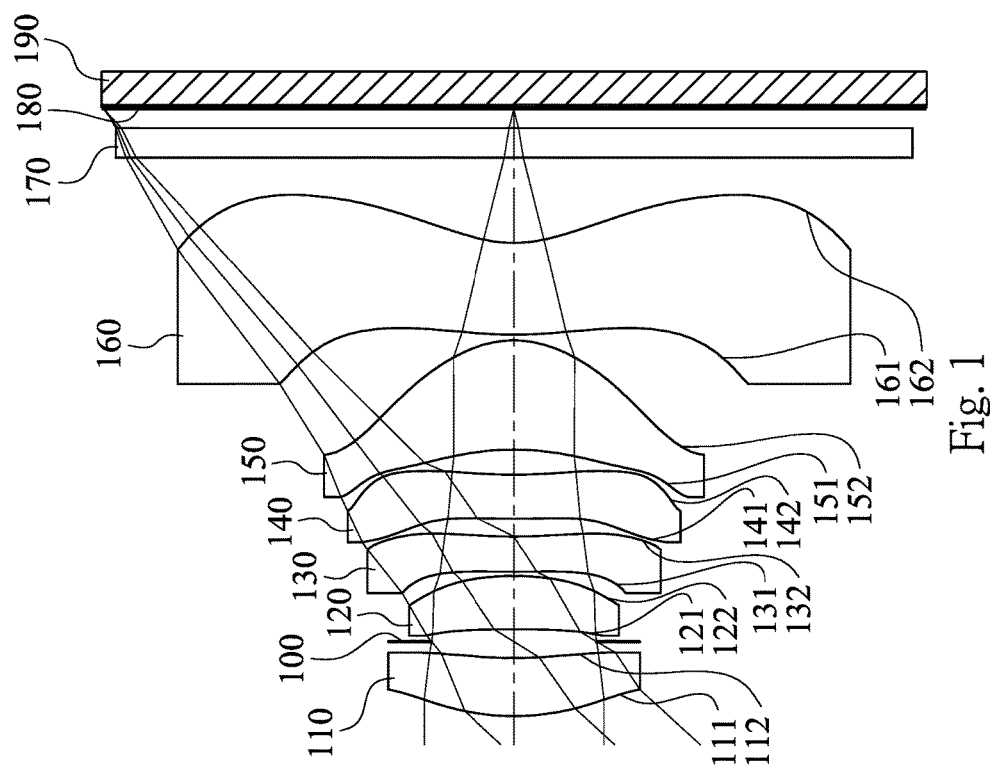


(56)

**References Cited**

U.S. PATENT DOCUMENTS

|              |    |         |             |
|--------------|----|---------|-------------|
| 2016/0124186 | A1 | 5/2016  | Tang et al. |
| 2016/0266354 | A1 | 9/2016  | Jang        |
| 2016/0341928 | A1 | 11/2016 | Liu et al.  |
| 2016/0341929 | A1 | 11/2016 | Liu et al.  |
| 2016/0341930 | A1 | 11/2016 | Liu et al.  |
| 2016/0341931 | A1 | 11/2016 | Liu et al.  |
| 2016/0341932 | A1 | 11/2016 | Liu et al.  |
| 2016/0341933 | A1 | 11/2016 | Liu et al.  |
| 2017/0227736 | A1 | 8/2017  | Lai et al.  |
| 2017/0227737 | A1 | 8/2017  | Lai et al.  |
| 2017/0227738 | A1 | 8/2017  | Lai et al.  |
| 2017/0227739 | A1 | 8/2017  | Lai et al.  |
| 2017/0235102 | A1 | 8/2017  | Lai et al.  |
| 2017/0235103 | A1 | 8/2017  | Lai et al.  |



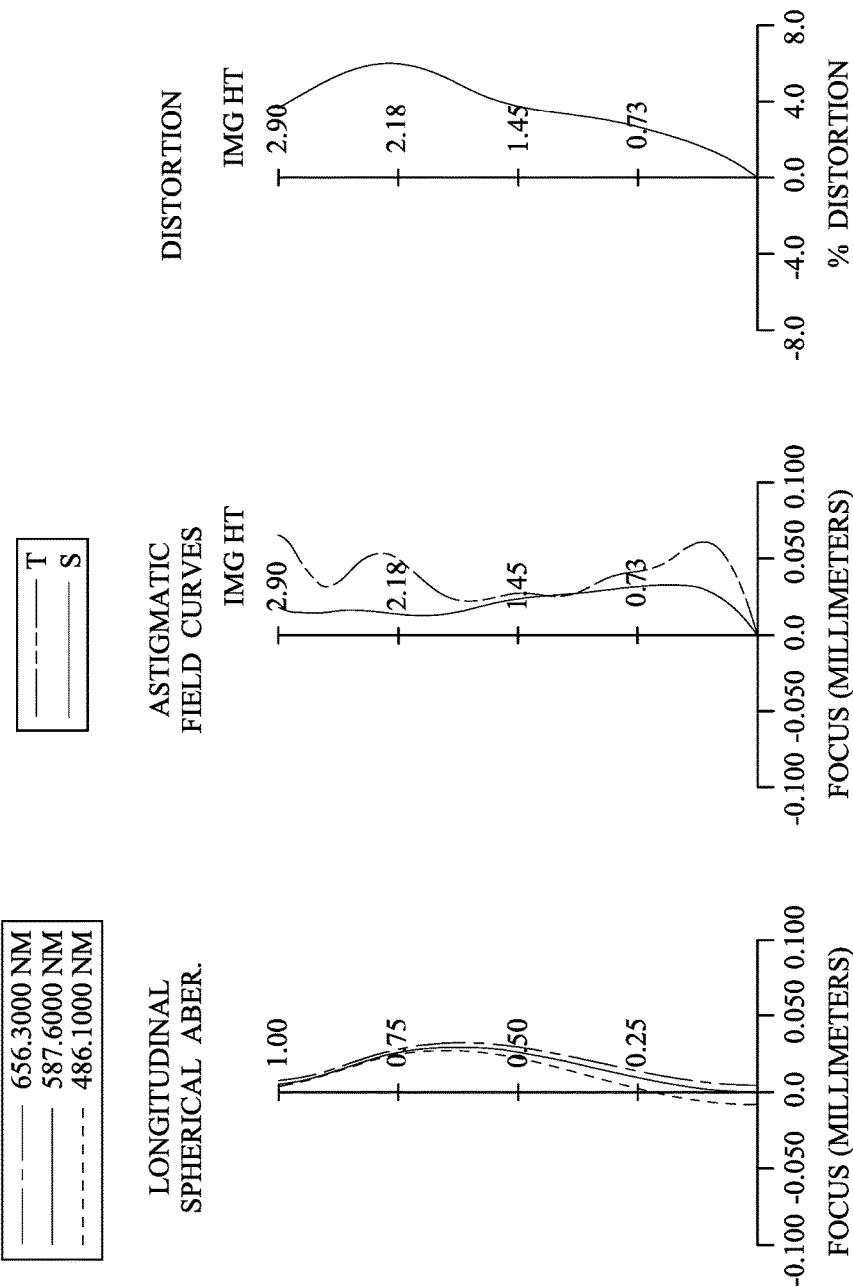


Fig. 2

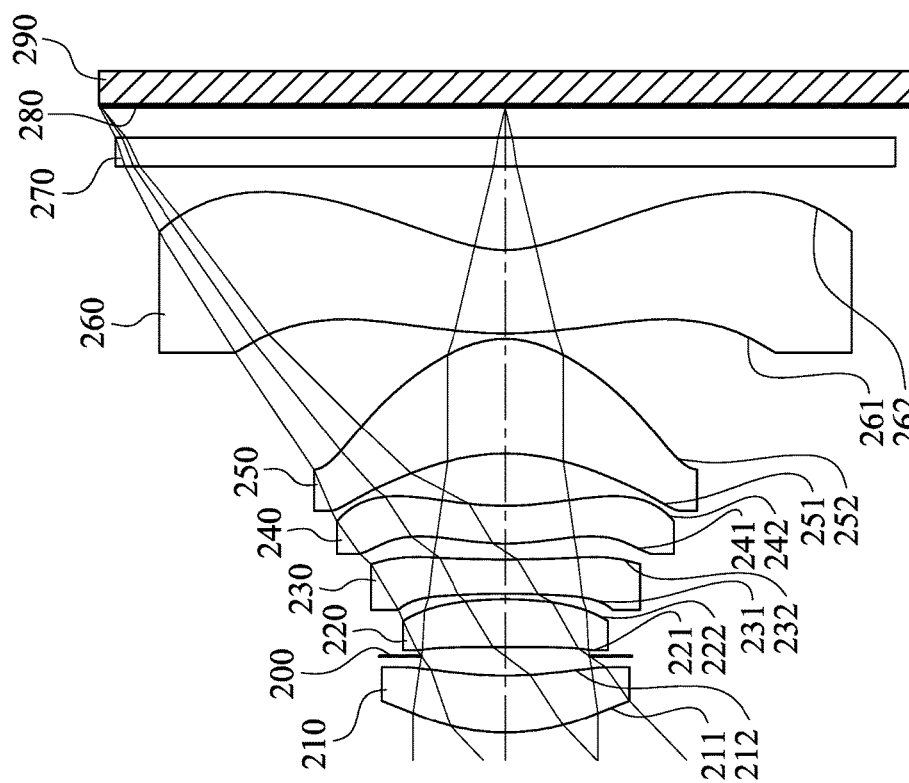


Fig. 3

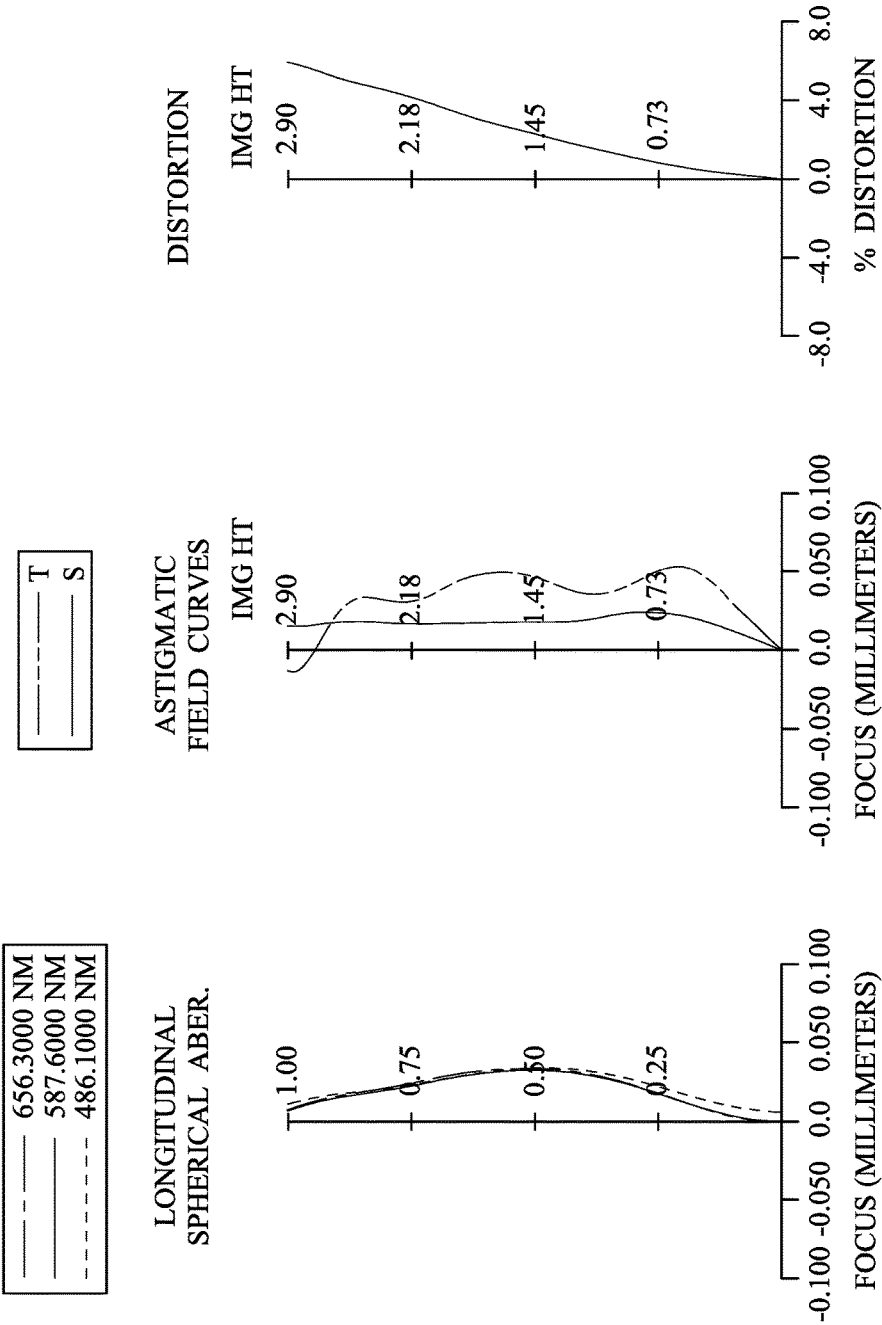


Fig. 4

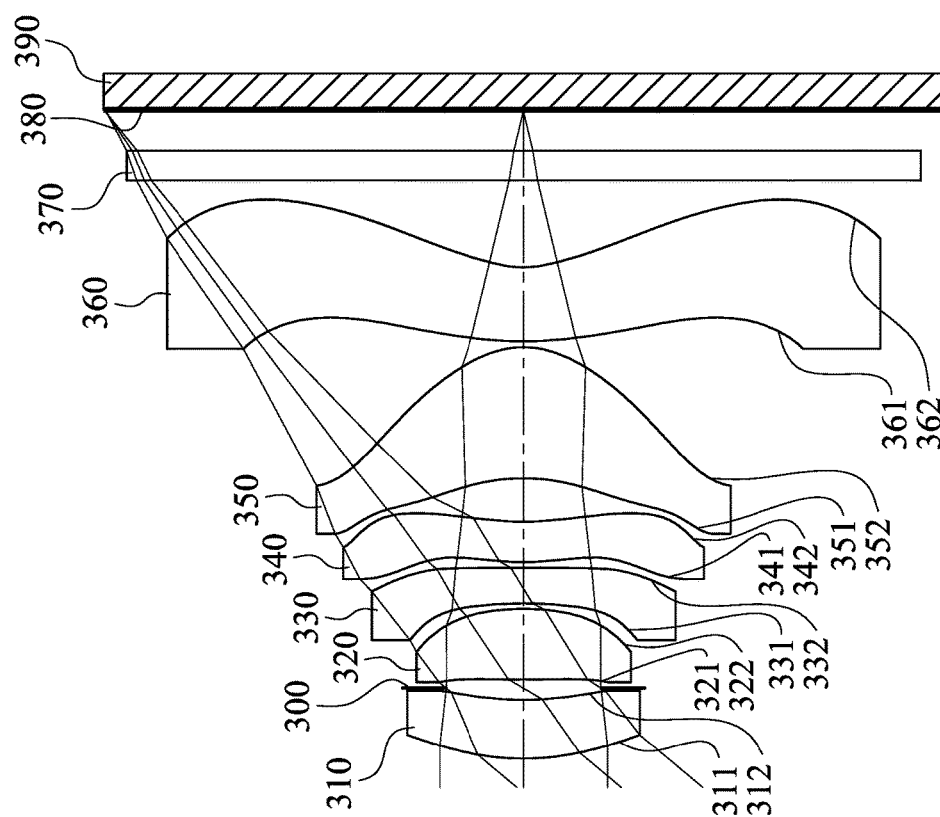
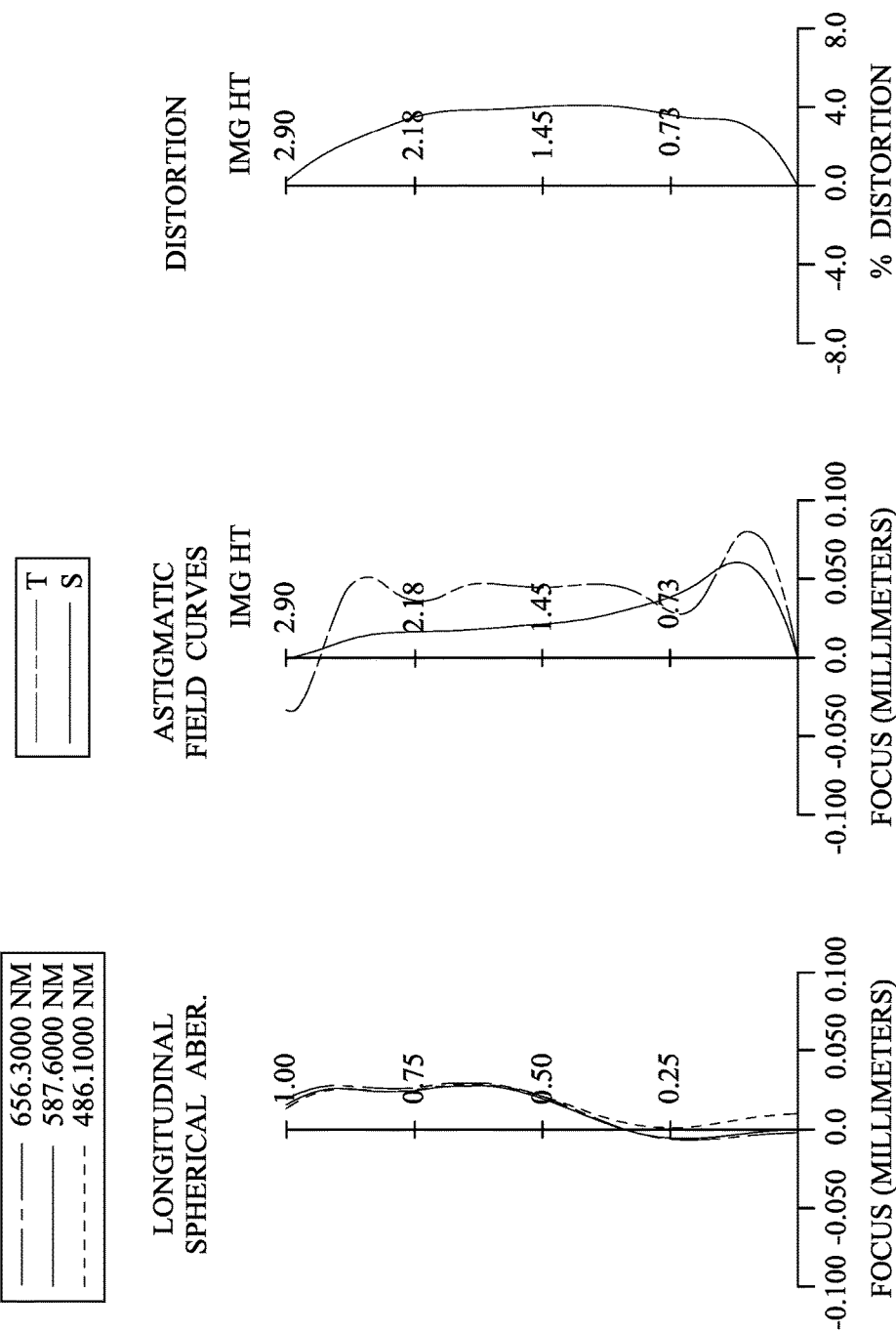


Fig. 5



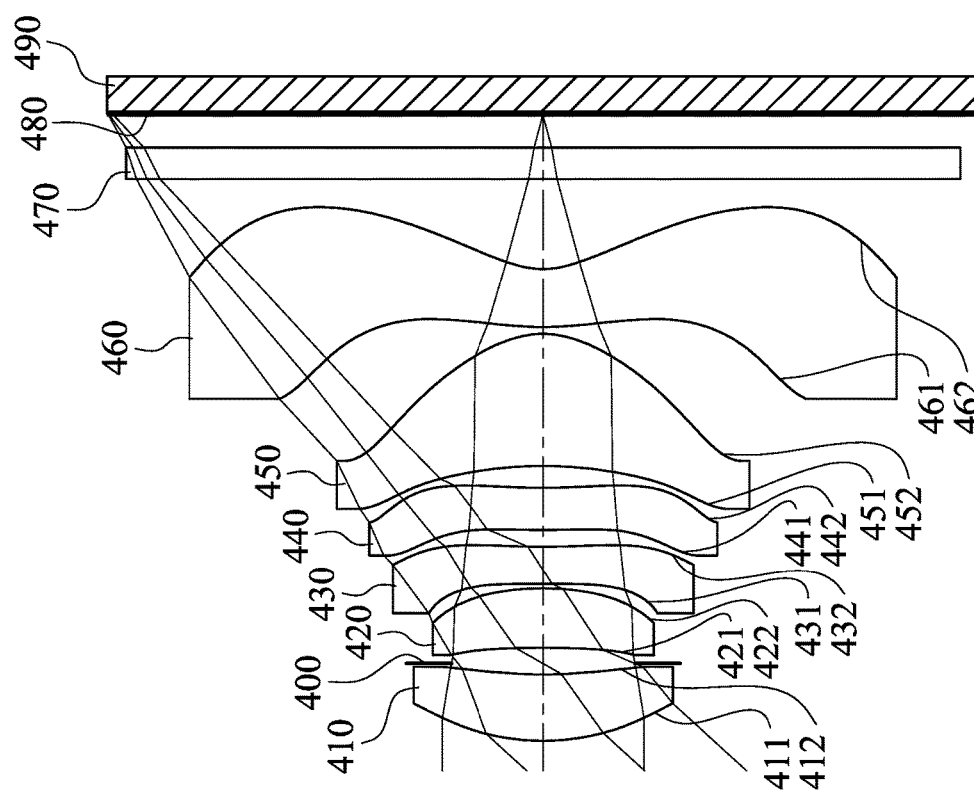


Fig. 7

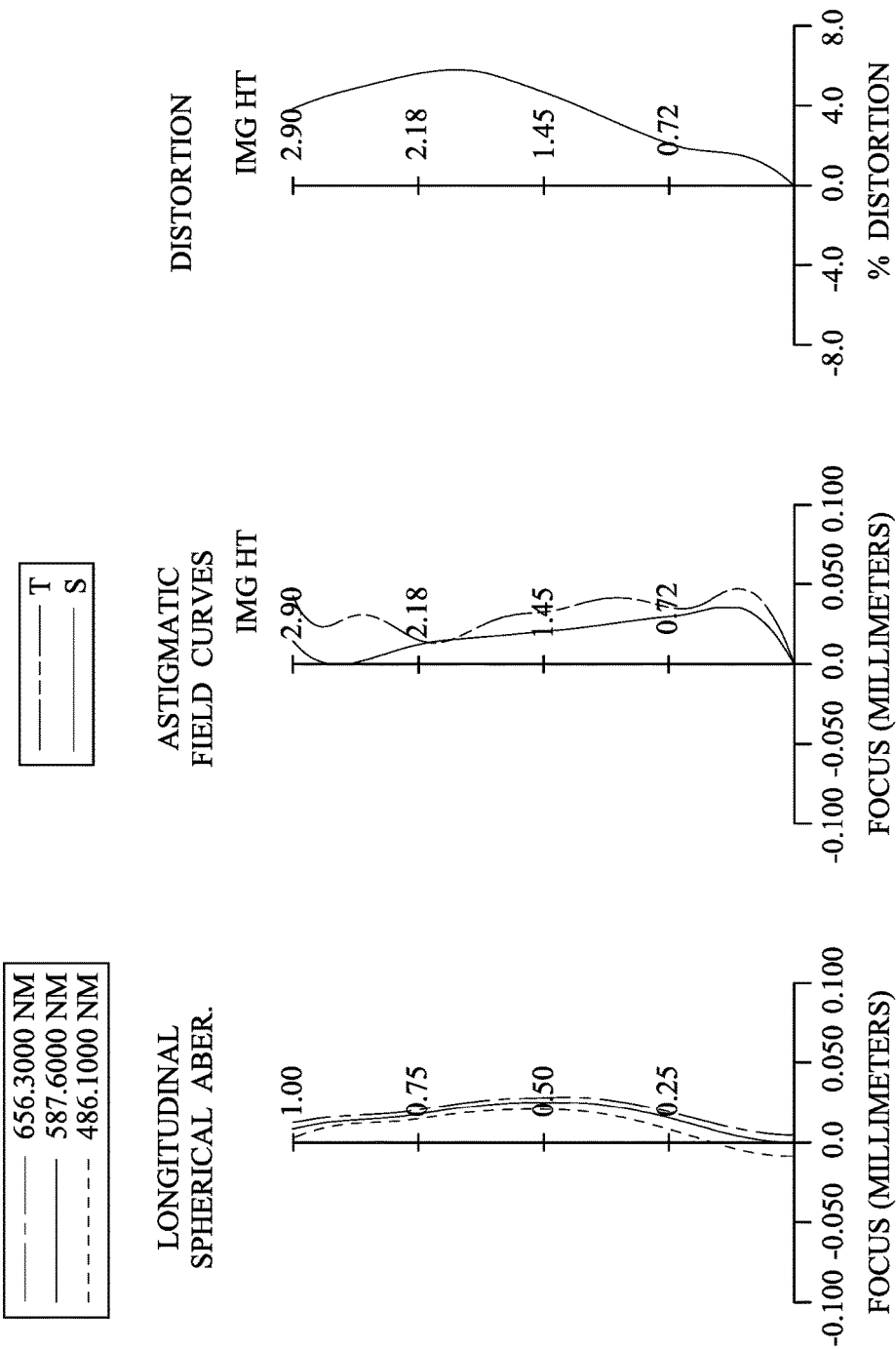


Fig. 8

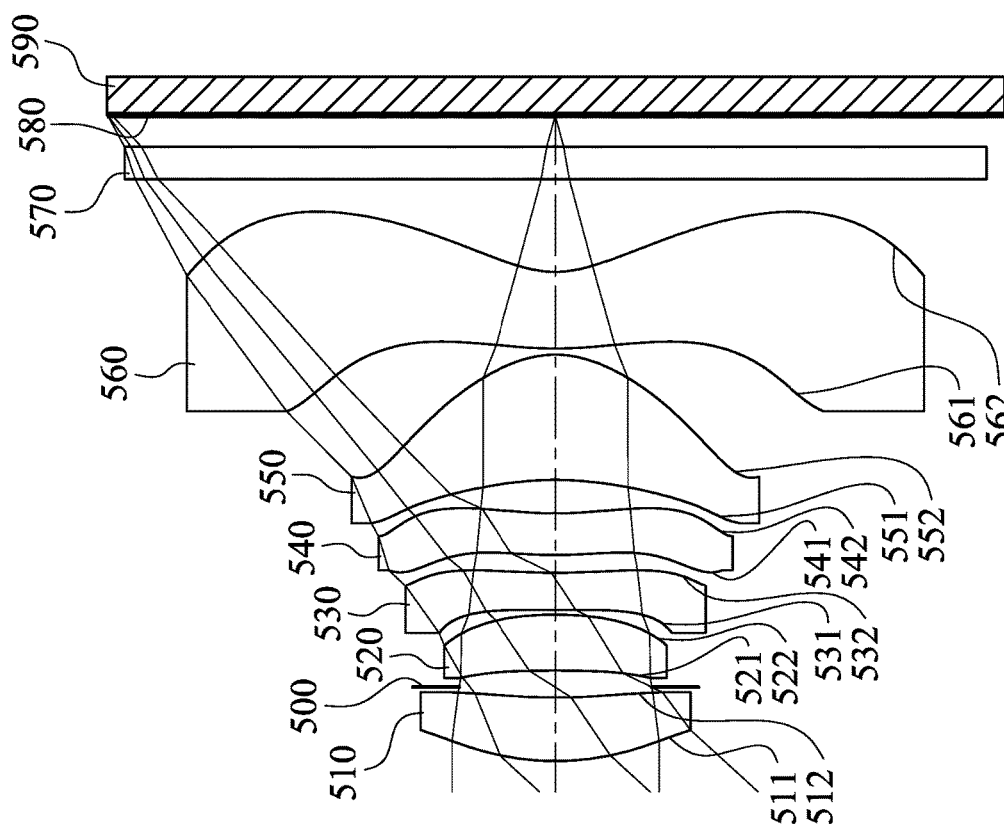
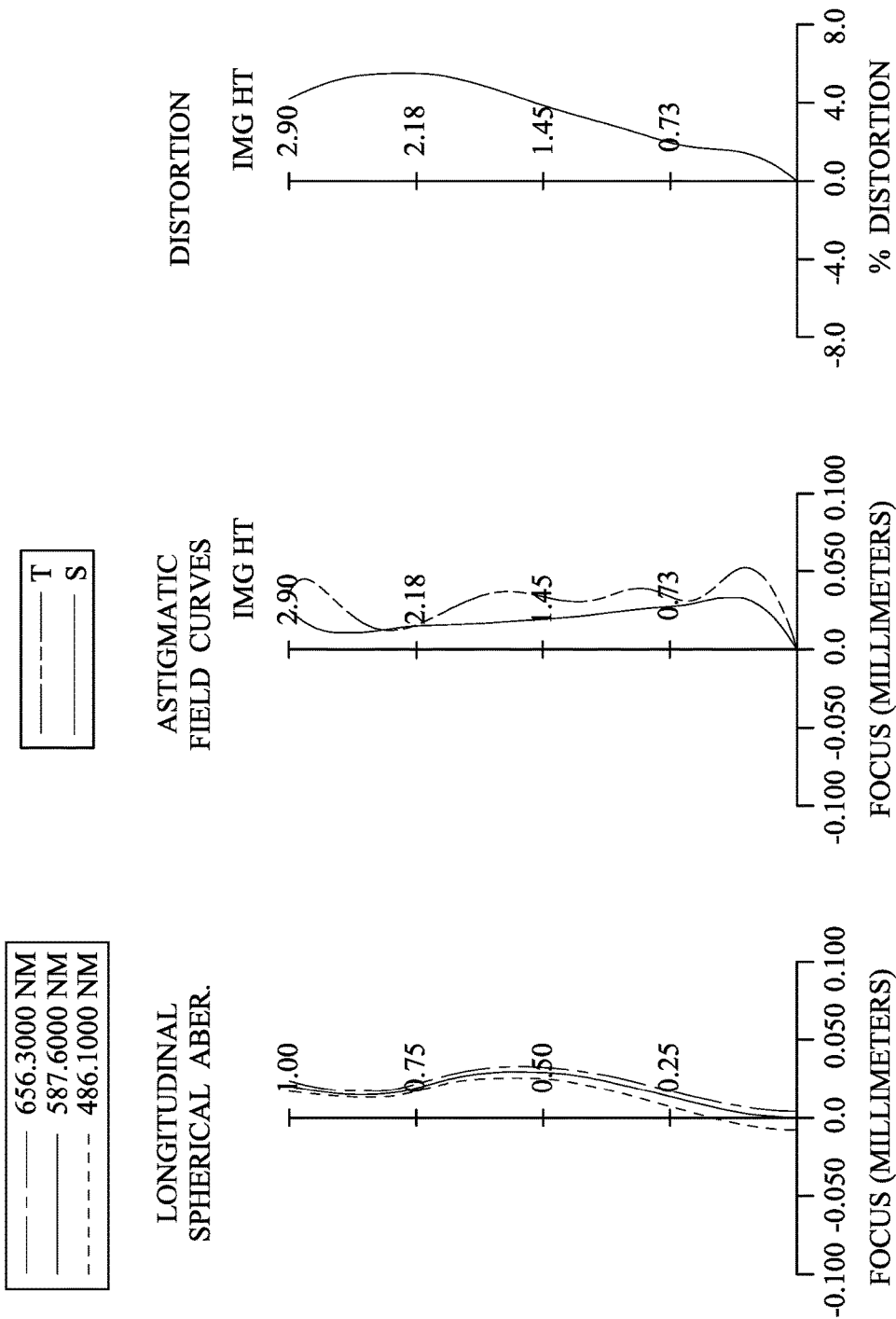


Fig. 9



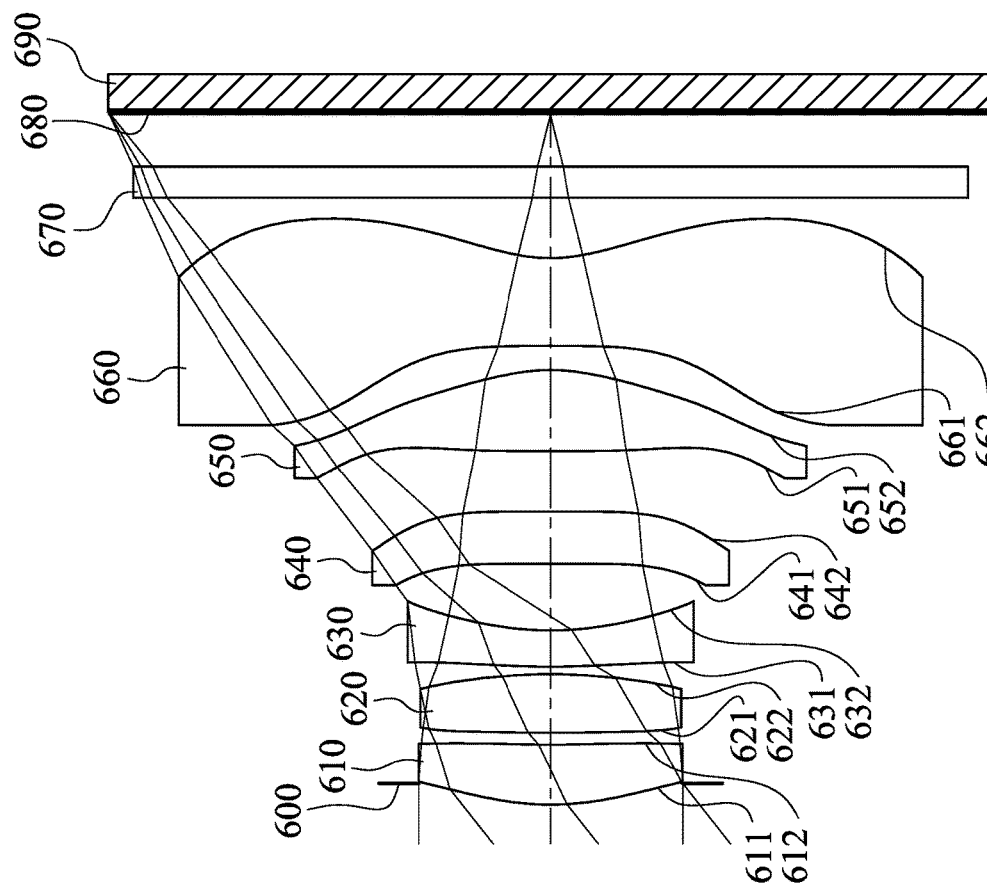


Fig. 11

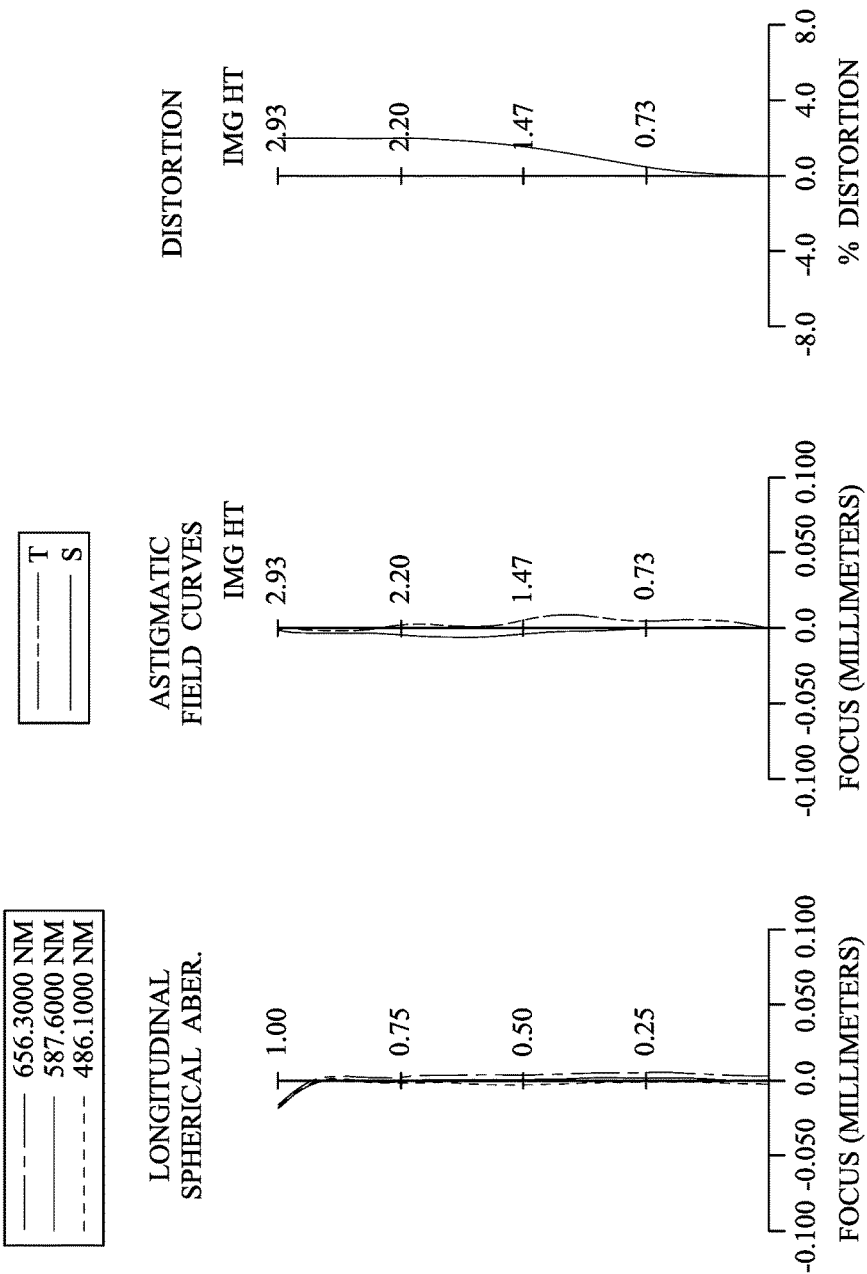
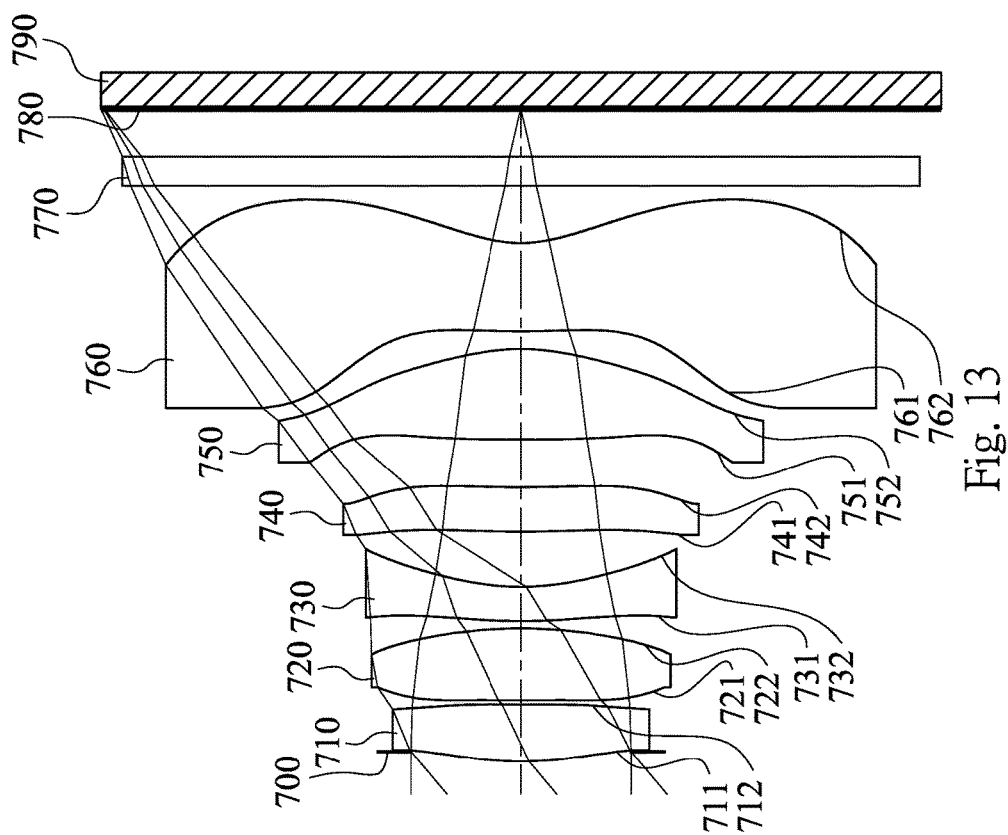


Fig. 12



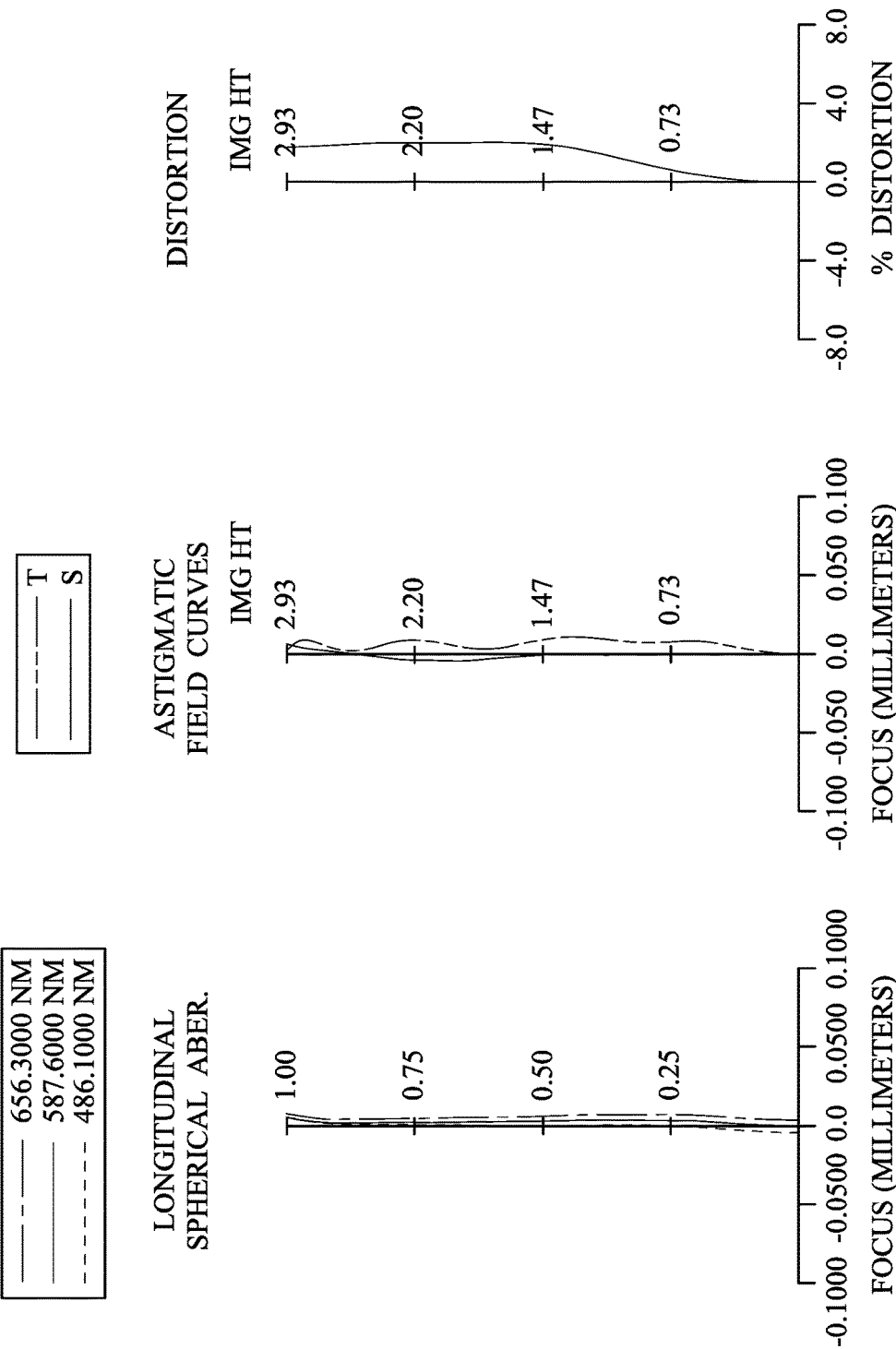


Fig. 14

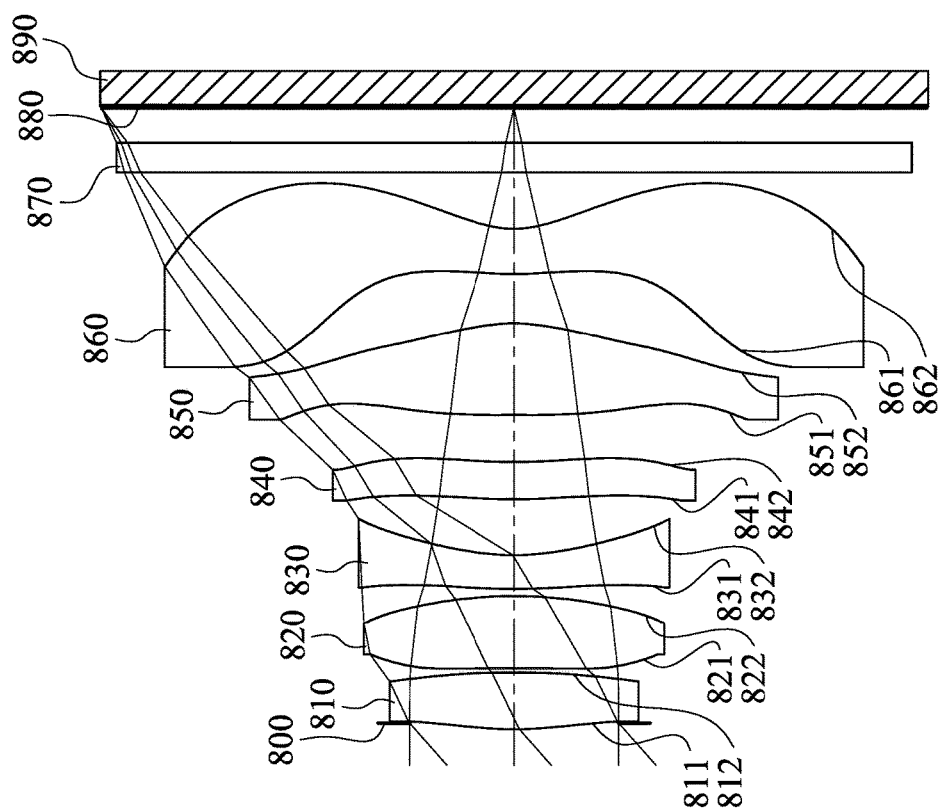
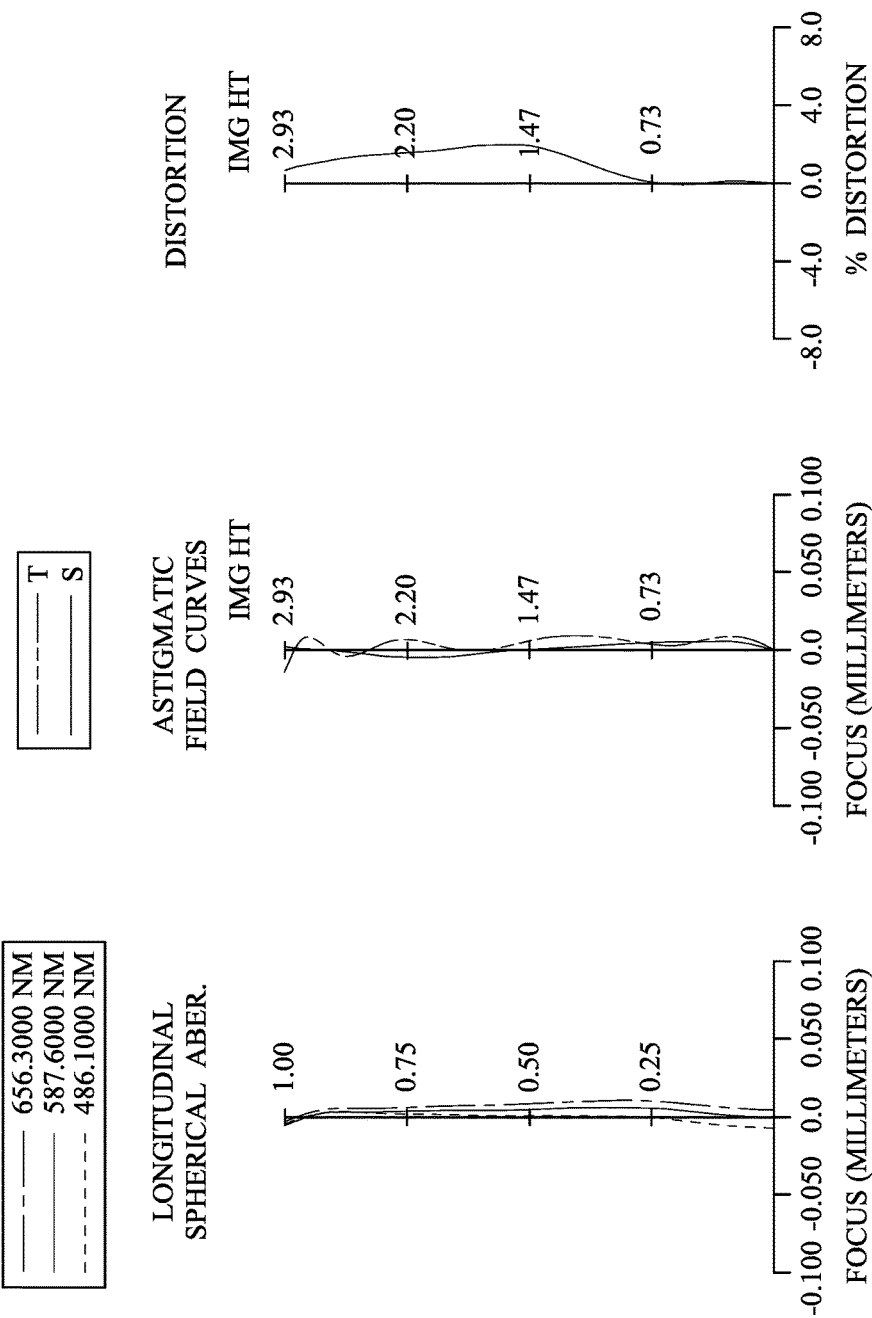


Fig. 15



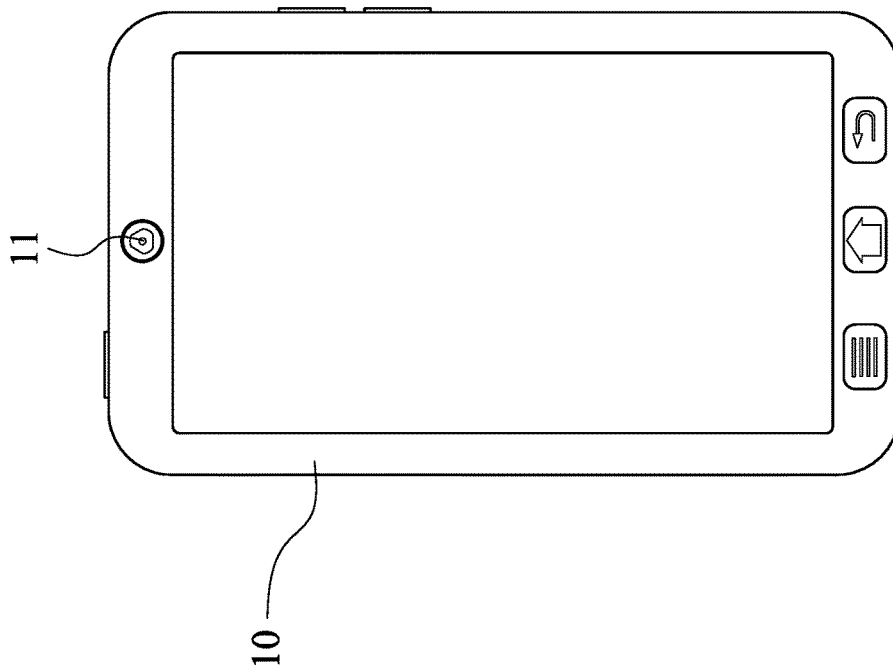


Fig. 17

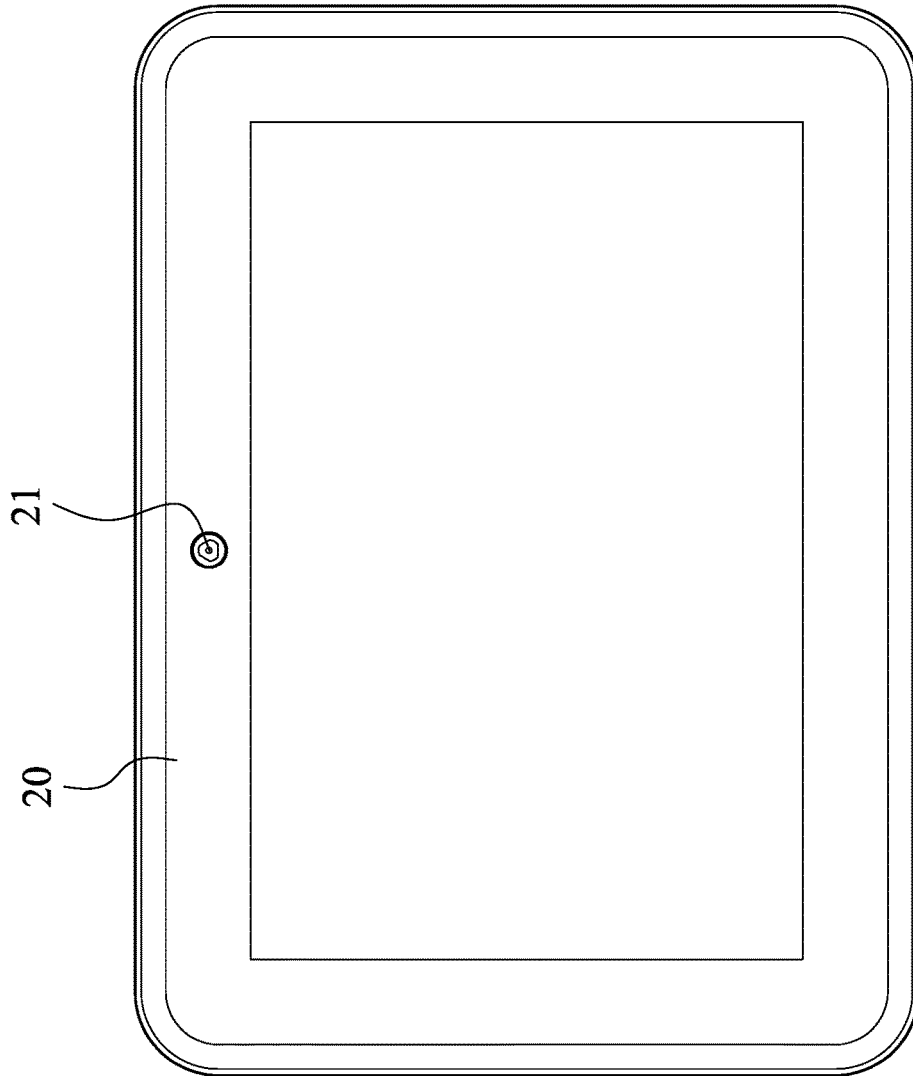


Fig. 18

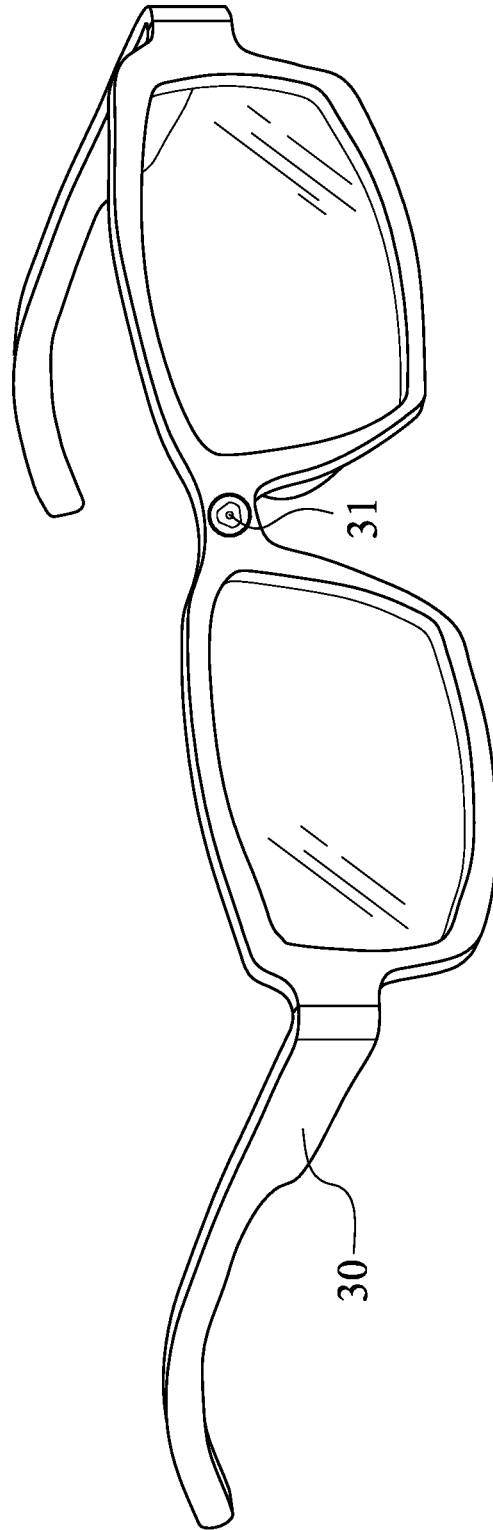


Fig. 19

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# OPTICAL IMAGE LENS ASSEMBLY, IMAGE CAPTURING DEVICE AND ELECTRONIC DEVICE

## RELATED APPLICATIONS

This application claims priority to Taiwan Application Serial Number 105118934, filed Jun. 16, 2016, which is herein incorporated by reference.

## BACKGROUND

### Technical Field

The present disclosure relates to an optical image lens assembly and an image capturing device. More particularly, the present disclosure relates to an optical image lens assembly and an image capturing device with a compact size applicable to electronic devices.

### Description of Related Art

In recent years, with the development of compact electronic devices based on popular demands, corresponding installed image capturing devices also advance toward trends of a large field of view, high image quality, miniaturization etc. However, conventional optical techniques cannot satisfy all trends at the same time. For examples, a lens assembly with a large field of view cannot be miniaturized due to the arrangement with spherical lens elements. A miniaturized lens assembly has insufficient aperture which would produce images with excessive noise in low light settings, lacking the depth of field effect, or unsatisfying results in popular applications due to insufficient field of view, etc. Hence, one of the goals in the optical lens industry is to find out how to achieve high image quality under the arrangement of large field of view, larger aperture and lens compactness at the same time.

## SUMMARY

According to one aspect of the present disclosure, an optical image lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element and a sixth lens element. The first lens element with positive refractive power has an object-side surface being convex in a paraxial region thereof. The second lens element with positive refractive power has an image-side surface being convex in a paraxial region thereof. The fourth lens element has an image-side surface being concave in a paraxial region thereof. The fifth lens element with positive refractive power has an object-side surface being concave in a paraxial region thereof and an image-side surface being convex in a paraxial region thereof, wherein the object-side surface and the image-side surface of the fifth lens element are aspheric. The sixth lens element with negative refractive power has an image-side surface being concave in a paraxial region thereof, wherein an object-side surface and the image-side surface of the sixth lens element are aspheric, and the image-side surface of the sixth lens element includes at least one convex shape in an off-axial region thereof. The optical image lens assembly has a total of six lens elements, and there is an air space between every two lens elements of the first lens element, the second lens element, the third lens element, the fourth lens element, the fifth lens element and the sixth lens element that are

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adjacent to each other. When a curvature radius of an object-side surface of the second lens element is  $R3$ , a curvature radius of the image-side surface of the second lens element is  $R4$ , a focal length of the second lens element is  $f2$ , a focal length of the third lens element is  $f3$ , and a focal length of the fourth lens element is  $f4$ , the following conditions are satisfied:

$$0.53 \leq (R3+R4)/(R3-R4);$$

$$|f2/f3| < 3.0; \text{ and}$$

$$|f3/f4| < 10.$$

According to another aspect of the present disclosure, an image capturing device includes the optical image lens assembly of the aforementioned aspect and an image sensor, wherein the image sensor is disposed on an image surface of the optical image lens assembly.

According to yet another aspect of the present disclosure, an electronic device includes the image capturing device of the aforementioned aspect.

According to further another aspect of the present disclosure, an optical image lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element and a sixth lens element. The first lens element with positive refractive power has an object-side surface being convex in a paraxial region thereof. The second lens element with positive refractive power has an image-side surface being convex in a paraxial region thereof. The third lens element with negative refractive power has an image-side surface being concave in a paraxial region thereof. The fifth lens element with positive refractive power has an object-side surface being concave in a paraxial region thereof and an image-side surface being convex in a paraxial region thereof, wherein the object-side surface and the image-side surface of the fifth lens element are aspheric. The sixth lens element with negative refractive power has an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof, wherein the object-side surface and the image-side surface of the sixth lens element are aspheric, and the image-side surface of the sixth lens element includes at least one convex shape in an off-axial region thereof. The optical image lens assembly has a total of six lens elements, and there is an air space between every two lens elements of the first lens element, the second lens element, the third lens element, the fourth lens element, the fifth lens element and the sixth lens element that are adjacent to each other. When a curvature radius of an object-side surface of the second lens element is  $R3$ , a curvature radius of the image-side surface of the second lens element is  $R4$ , a focal length of the second lens element is  $f2$ , and a focal length of the third lens element is  $f3$ , the following conditions are satisfied:

$$0.53 \leq (R3+R4)/(R3-R4), \text{ and}$$

$$|f2/f3| < 3.0.$$

## BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic view of an image capturing device according to the 1st embodiment of the present disclosure;

FIG. 2 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 1st embodiment;

FIG. 3 is a schematic view of an image capturing device according to the 2nd embodiment of the present disclosure;

FIG. 4 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 2nd embodiment;

FIG. 5 is a schematic view of an image capturing device according to the 3rd embodiment of the present disclosure;

FIG. 6 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 3rd embodiment;

FIG. 7 is a schematic view of an image capturing device according to the 4th embodiment of the present disclosure;

FIG. 8 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 4th embodiment;

FIG. 9 is a schematic view of an image capturing device according to the 5th embodiment of the present disclosure;

FIG. 10 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 5th embodiment;

FIG. 11 is a schematic view of an image capturing device according to the 6th embodiment of the present disclosure;

FIG. 12 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 6th embodiment;

FIG. 13 is a schematic view of an image capturing device according to the 7th embodiment of the present disclosure;

FIG. 14 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 7th embodiment;

FIG. 15 is a schematic view of an image capturing device according to the 8th embodiment of the present disclosure;

FIG. 16 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 8th embodiment;

FIG. 17 is a schematic view of an electronic device according to the 9th embodiment of the present disclosure;

FIG. 18 is a schematic view of an electronic device according to the 10th embodiment of the present disclosure; and

FIG. 19 is a schematic view of an electronic device according to the 11th embodiment of the present disclosure.

#### DETAILED DESCRIPTION

An optical image lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element and a sixth lens element. The optical image lens assembly has a total of six lens elements.

According to the optical image lens assembly of the present disclosure, there is an air space between every two lens elements of the first lens element, the second lens element, the third lens element, the fourth lens element, the fifth lens element and the sixth lens element that are adjacent to each other. That is, each of the first through sixth lens elements is a single and non-cemented lens element, and there is a space between every two adjacent lens elements. Moreover, the manufacturing process of the cemented lenses is more complex than the non-cemented lenses. In particular, a cementing surface of one lens element and a cementing surface of the following lens element need to have accurate curvature to ensure these two lens elements will be highly cemented. However, during the cementing process, those two lens elements might not be highly cemented due to displacements and it is thereby not favorable for image quality of the optical image lens assembly. Therefore, according to the optical image lens assembly of the present disclosure, having an air space in a paraxial region between

every two adjacent lens elements avoids the problem generated by the cemented lens elements.

The first lens element with positive refractive power has an object-side surface being convex in a paraxial region thereof, and can have an image-side surface being concave in a paraxial region thereof. Therefore, the main light converging ability can be provided so as to reduce the volume of the optical image lens assembly and obtain an optimal compact size thereof.

The second lens element with positive refractive power has an image-side surface being convex in a paraxial region thereof. By such arrangement, the proper amount of positive refractive power of the first lens element can be provided, so that it is favorable for avoiding aberrations from excessive refraction of the off-axial light with a large aperture or low tolerance of the optical image lens assembly (against manufacturing errors, such as decentering, of the first lens element).

The third lens element can have negative refractive power, and can have an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof. Therefore, it is favorable for correcting aberrations generated from the first lens element and the second lens element. Furthermore, the image-side surface of the third lens element can include at least one convex shape in an off-axial region thereof. Therefore, it is favorable for enhancing the image quality by correcting aberrations of the off-axial region on the image.

The fourth lens element can have an image-side surface being concave in a paraxial region thereof, so that astigmatism of the optical image lens assembly can be corrected. Furthermore, the image-side surface of the fourth lens element can include at least one convex shape in an off-axial region thereof. Thus, it is favorable for enhancing the image quality by correcting aberrations of the off-axial region on the image.

The fifth lens element with positive refractive power has an object-side surface being concave in a paraxial region thereof and an image-side surface being convex in a paraxial region thereof. Therefore, astigmatism of the optical image lens assembly can be corrected by the meniscus lens arrangement while reducing the sensitivity thereof.

The sixth lens element with negative refractive power can have an object-side surface being convex in a paraxial region thereof, and has an image-side surface being concave in a paraxial region thereof. Therefore, the principal point can be positioned closer to the object side, so that the back focal length can be reduced for lens miniaturization with effective space utilization of the optical image lens assembly adopted in the camera module. Also, the image-side surface of the sixth lens element includes at least one convex shape in an off-axial region thereof, so that off-axial aberrations can be corrected.

When a curvature radius of the object-side surface of the second lens element is  $R3$ , and a curvature radius of the image-side surface of the second lens element is  $R4$ , the following condition is satisfied:  $0.53 \leq (R3+R4)/(R3-R4)$ . Therefore, it is favorable for positioning the principal point of the second lens element closer to the image surface for enlarging the field of view of the optical image lens assembly and correcting aberrations. Preferably, the following condition can be satisfied:  $0.80 < (R3+R4)/(R3-R4) < 4.0$ . More preferably, the following condition can be satisfied:  $0.90 < (R3+R4)/(R3-R4) < 3.50$ .

When a focal length of the second lens element is  $f2$ , and a focal length of the third lens element is  $f3$ , the following condition is satisfied:  $|f2/f3| < 3.0$ . Therefore, the positive

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refractive power of the second lens element will be sufficient so as to provide the functionality with the corresponding first lens element with positive refractive power. More preferably, the following condition can be satisfied:  $0 < |f_2/f_3| < 1.50$ .

When the focal length of the third lens element is  $f_3$ , and a focal length of the fourth lens element is  $f_4$ , the following condition is satisfied:  $|f_3/f_4| < 10$ . Therefore, it is favorable for correcting aberrations and enhancing the image quality. More preferably, the following condition can be satisfied:  $|f_3/f_4| < 2.0$ .

When a curvature radius of the object-side surface of the first lens element is  $R_1$ , and a curvature radius of the image-side surface of the first lens element is  $R_2$ , the following condition is satisfied:  $(R_1+R_2)/(R_1-R_2) < -1.60$ . Therefore, it is favorable for avoiding the excessive positive refractive power of the first lens element so as to provide features and effects thereof with the corresponding second lens element.

When an axial distance between the object-side surface of the first lens element and the image-side surface of the sixth lens element is  $T_d$ , and a sum of axial distances between every two of the lens elements of the optical image lens assembly that are adjacent to each other is  $\Sigma AT$ , the following condition is satisfied:  $4.0 < T_d/\Sigma AT < 20$ . Therefore, it is favorable for utilizing the space of the optical image lens assembly by closely arranging the lens elements thereof.

When a focal length of the first lens element is  $f_1$ , and the focal length of the second lens element is  $f_2$ , the following condition is satisfied:  $0.50 < f_1/f_2 < 2.0$ . Therefore, it is favorable for providing the functionality of both the first lens element and the second lens element having positive refractive power for reducing the sensitivity. More preferably, the following condition can be satisfied:  $0.50 < f_1/f_2 < 1.50$ .

When a curvature radius of the object-side surface of the fifth lens element is  $R_9$ , and a curvature radius of the image-side surface of the fifth lens element is  $R_{10}$ , and the following condition is satisfied:  $0 < (R_9-R_{10})/(R_9+R_{10}) < 0.75$ . Therefore, it is favorable for correcting aberrations and astigmatism by the meniscus shape configuration of the fifth lens element.

When the focal length of the first lens element is  $f_1$ , the focal length of the second lens element is  $f_2$ , the focal length of the third lens element is  $f_3$ , the focal length of the fourth lens element is  $f_4$ , a focal length of the fifth lens element is  $f_5$ , and a focal length of the sixth lens element is  $f_6$ , the following conditions are satisfied:  $|f_1| < |f_4|$ ,  $|f_2| < |f_4|$ ,  $|f_3| < |f_4|$ ,  $|f_5| < |f_4|$ , and  $|f_6| < |f_4|$ . Therefore, high order aberrations can be corrected by arranging the fourth lens element with weaker refractive power in the optical image lens assembly (that is, the fourth lens element has the weakest refractive power among the lens elements), so that the better image quality can be obtained.

When a central thickness of the second lens element is  $CT_2$ , and an axial distance between the first lens element and the second lens element is  $T_{12}$ , the following condition is satisfied:  $0 < CT_2/T_{12} < 5.0$ . Therefore, the interference between the lens elements can be avoided in lens assembling by having sufficient space between the first lens element and the second lens element, and it is also favorable for arranging other elements, such as a light blocking sheet, between the first lens element and the second lens element.

When an Abbe number of the second lens element is  $V_2$ , and an Abbe number of the third lens element is  $V_3$ , the following condition is satisfied:  $0.20 < V_3/V_2 < 0.60$ . Therefore, it is favorable for balancing corrections of chromatic aberration and astigmatism.

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When a focal length of the optical image lens assembly is  $f$ , a curvature radius of the object-side surface of the first lens element is  $R_1$ , and a curvature radius of the image-side surface of the first lens element is  $R_2$ , the following condition is satisfied:  $1.0 < f/R_1 + f/R_2 < 5.0$ . Therefore, it is favorable for avoiding the surfaces of the first lens element being overly flat with oversized effective radii thereof and difficult for miniaturization.

When the focal length of the optical image lens assembly is  $f$ , and a curvature radius of the image-side surface of the fourth lens element is  $R_8$ , the following condition is satisfied:  $0 < R_8/f < 2.5$ . Therefore, it is favorable for avoiding overly curved image-side surface of the fourth lens element and stray light generated from unwanted surface reflections due to excessive curvature.

When a maximum image height of the optical image lens assembly is  $ImgH$ , and the focal length of the optical image lens assembly is  $f$ , the following condition is satisfied:  $0.85 < ImgH/f < 2.0$ . Therefore, it is favorable for providing the optical image lens assembly with a large field of view.

When the Abbe number of the third lens element is  $V_3$ , an Abbe number of the fifth lens element is  $V_5$ , and an Abbe number of the sixth lens element is  $V_6$ , the following condition is satisfied:  $0.50 < (V_3+V_6)/V_5 < 1.20$ . Therefore, it is favorable for balancing corrections of chromatic aberration and astigmatism.

According to the optical image lens assembly of the present disclosure, the lens elements thereof can be made of glass or plastic materials. When the lens elements are made of glass materials, the distribution of the refractive power of the optical image lens assembly may be more flexible to design. When the lens elements are made of plastic materials, manufacturing costs can be effectively reduced. Furthermore, surfaces of each lens element can be arranged to be aspheric, since the aspheric surface of the lens element is easy to form a shape other than a spherical surface so as to have more controllable variables for eliminating aberrations thereof, and to further decrease the required amount of lens elements in the optical image lens assembly. Therefore, the total track length of the optical image lens assembly can also be reduced.

According to the optical image lens assembly of the present disclosure, each of an object-side surface and an image-side surface has a paraxial region and an off-axial region. The paraxial region refers to the region of the surface where light rays travel close to an optical axis, and the off-axial region refers to the region of the surface away from the paraxial region. Particularly, when the lens element has a convex surface, it indicates that the surface can be convex in the paraxial region thereof; when the lens element has a concave surface, it indicates that the surface can be concave in the paraxial region thereof.

According to the optical image lens assembly of the present disclosure, the refractive power or the focal length of a lens element being positive or negative may refer to the refractive power or the focal length in a paraxial region of the lens element.

According to the optical image lens assembly of the present disclosure, the optical image lens assembly can include at least one stop, such as an aperture stop, a glare stop or a field stop. Said glare stop or said field stop is for eliminating the stray light and thereby improving the image resolution thereof.

According to the optical image lens assembly of the present disclosure, the image surface of the optical image lens assembly, based on the corresponding image sensor, can

be flat or curved. In particular, the image surface can be a curved surface being concave facing towards the object side.

According to the optical image lens assembly of the present disclosure, an aperture stop can be configured as a front stop or a middle stop. A front stop disposed between an object and the first lens element can provide a longer distance between an exit pupil of the optical image lens assembly and the image surface, and thereby obtains a telecentric effect and improves the image-sensing efficiency of the image sensor, such as CCD or CMOS. A middle stop disposed between the first lens element and the image surface is favorable for enlarging the field of view of the optical image lens assembly and thereby provides a wider field of view for the same.

According to the optical image lens assembly of the present disclosure, the optical image lens assembly can be applied to 3D (three-dimensional) image capturing applications, in products such as digital cameras, mobile devices, digital tablets, smart TVs, surveillance systems, motion sensing input devices, driving recording systems, rearview camera systems, and wearable devices.

According to the present disclosure, an image capturing device is provided. The image capturing device includes the aforementioned optical image lens assembly and an image sensor, wherein the image sensor is disposed on the image side of the aforementioned optical image lens assembly, that is, the image sensor can be disposed on or near the image surface of the aforementioned optical image lens assembly. By the arrangement of positive refractive power of both of the first lens element and the second lens element which are closest to the imaged object, the excessive positive refractive power of the first lens element can be avoided, so that it is favorable for avoiding excessive aberrations or over sensitivity of the tolerance (such as the eccentricity) of the first lens element which are generated by the over refraction of the off-axial light under large aperture. Thus, the excessive positive refractive power of the first lens element generated from the conventional arrangement of the first lens element with positive refractive power and the second lens element with negative refractive power can be improved. Preferably, the image capturing device can further include a barrel member, a holder member or a combination thereof.

According to the present disclosure, an electronic device is provided, which includes the aforementioned image capturing device. Therefore, image quality of the electronic device can be improved. Preferably, the electronic device can further include but not limited to a control unit, a display, a storage unit, a random access memory unit (RAM) or a combination thereof.

According to the above description of the present disclosure, the following 1st-11th specific embodiments are provided for further explanation.

#### 1st Embodiment

FIG. 1 is a schematic view of an image capturing device according to the 1st embodiment of the present disclosure. FIG. 2 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 1st embodiment. In FIG. 1, the image capturing device includes an optical image lens assembly (its reference numeral is omitted) and an image sensor 190. The optical image lens assembly includes, in order from an object side to an image side, a first lens element 110, an aperture stop 100, a second lens element 120, a third lens element 130, a fourth lens element 140, a fifth lens element 150, a sixth lens element 160, an IR-cut filter 170 and an

image surface 180, wherein the image sensor 190 is disposed on the image surface 180 of the optical image lens assembly. The optical image lens assembly has a total of six lens elements (110-160), and there is an air space between every two lens elements of the first lens element 110, the second lens element 120, the third lens element 130, the fourth lens element 140, the fifth lens element 150 and the sixth lens element 160 that are adjacent to each other.

The first lens element 110 with positive refractive power has an object-side surface 111 being convex in a paraxial region thereof and an image-side surface 112 being concave in a paraxial region thereof. The first lens element 110 is made of a plastic material, and has the object-side surface 111 and the image-side surface 112 being both aspheric.

The second lens element 120 with positive refractive power has an object-side surface 121 being concave in a paraxial region thereof and an image-side surface 122 being convex in a paraxial region thereof. The second lens element 120 is made of a plastic material, and has the object-side surface 121 and the image-side surface 122 being both aspheric.

The third lens element 130 with negative refractive power has an object-side surface 131 being convex in a paraxial region thereof and an image-side surface 132 being concave in a paraxial region thereof. The third lens element 130 is made of a plastic material, and has the object-side surface 131 and the image-side surface 132 being both aspheric. Furthermore, the image-side surface 132 of the third lens element 130 includes at least one convex shape in an off-axial region thereof.

The fourth lens element 140 with negative refractive power has an object-side surface 141 being convex in a paraxial region thereof and an image-side surface 142 being concave in a paraxial region thereof. The fourth lens element 140 is made of a plastic material, and has the object-side surface 141 and the image-side surface 142 being both aspheric. Furthermore, the image-side surface 142 of the fourth lens element 140 includes at least one convex shape in an off-axial region thereof.

The fifth lens element 150 with positive refractive power has an object-side surface 151 being concave in a paraxial region thereof and an image-side surface 152 being convex in a paraxial region thereof. The fifth lens element 150 is made of a plastic material, and has the object-side surface 151 and the image-side surface 152 being both aspheric.

The sixth lens element 160 with negative refractive power has an object-side surface 161 being convex in a paraxial region thereof and an image-side surface 162 being concave in a paraxial region thereof. The sixth lens element 160 is made of a plastic material, and has the object-side surface 161 and the image-side surface 162 being both aspheric. Furthermore, the image-side surface 162 of the sixth lens element 160 includes at least one convex shape in an off-axial region thereof.

The IR-cut filter 170 is made of a glass material and located between the sixth lens element 160 and the image surface 180, and will not affect the focal length of the optical image lens assembly.

The equation of the aspheric surface profiles of the aforementioned lens elements of the 1st embodiment is expressed as follows:

$$X(Y) = (Y^2/R)/(1 - \sqrt{1 - (1+k) \times (Y/R)^2}) + \sum_i (Ai) \times (Y^i),$$

where,

X is the relative distance between a point on the aspheric surface spaced at a distance Y from the optical axis and the tangential plane at the aspheric surface vertex on the optical axis;

Y is the vertical distance from the point on the aspheric surface to the optical axis;

R is the curvature radius;

k is the conic coefficient; and

Ai is the i-th aspheric coefficient.

In the optical image lens assembly according to the 1st embodiment, when a focal length of the optical image lens assembly is f, an f-number of the optical image lens assembly is Fno, and half of a maximum field of view of the optical image lens assembly is HFOV, these parameters have the following values: f=2.55 mm; Fno=2.01; and HFOV=47.5 degrees.

In the optical image lens assembly according to the 1st embodiment, when an Abbe number of the second lens element 120 is V2, an Abbe number of the third lens element 130 is V3, an Abbe number of the fifth lens element 150 is V5, and an Abbe number of the sixth lens element 160 is V6, the following conditions are satisfied:  $V3/V2=0.36$ ; and  $(V3+V6)/V5=0.79$ .

In the optical image lens assembly according to the 1st embodiment, when a central thickness of the second lens element 120 is CT2, and an axial distance between the first lens element 110 and the second lens element 120 is T12, the following condition is satisfied:  $CT2/T12=1.87$ .

In the optical image lens assembly according to the 1st embodiment, when an axial distance between the object-side surface 111 of the first lens element 110 and the image-side surface 162 of the sixth lens element 160 is Td, the axial distance between the first lens element 110 and the second lens element 120 is T12, an axial distance between the second lens element 120 and the third lens element 130 is T23, an axial distance between the third lens element 130 and the fourth lens element 140 is T34, an axial distance between the fourth lens element 140 and the fifth lens element 150 is T45, an axial distance between the fifth lens element 150 and the sixth lens element 160 is T56, and a sum of axial distances between every two of the lens elements of the optical image lens assembly that are adjacent to each other is  $\Sigma AT$  (that is,  $\Sigma AT=T12+T23+T34+T45+T56$ ), the following condition is satisfied:  $Td/\Sigma AT=5.84$ .

In the optical image lens assembly according to the 1st embodiment, when a maximum image height of the optical image lens assembly is ImgH (half of a diagonal length of an effective photosensitive area of the image sensor 190), and the focal length of the optical image lens assembly is f, the following condition is satisfied:  $ImgH/f=1.14$ .

In the optical image lens assembly according to the 1st embodiment, when a curvature radius of the object-side surface 111 of the first lens element 110 is R1, and a curvature radius of the image-side surface 112 of the first lens element 110 is R2, the following condition is satisfied:  $(R1+R2)/(R1-R2)=-2.88$ .

In the optical image lens assembly according to the 1st embodiment, when a curvature radius of the object-side surface 121 of the second lens element 120 is R3, and a curvature radius of the image-side surface 122 of the second lens element 120 is R4, the following condition is satisfied:  $(R3+R4)/(R3-R4)=1.51$ .

In the optical image lens assembly according to the 1st embodiment, when a curvature radius of the object-side surface 151 of the fifth lens element 150 is R9, and a curvature radius of the image-side surface 152 of the fifth lens element 150 is R10, the following condition is satisfied:  $(R9-R10)/(R9+R10)=0.47$ .

In the optical image lens assembly according to the 1st embodiment, when the focal length of the optical image lens assembly is f, the curvature radius of the object-side surface 111 of the first lens element 110 is R1, and the curvature radius of the image-side surface 112 of the first lens element 110 is R2, the following condition is satisfied:  $f/R1+f/R2=2.22$ .

In the optical image lens assembly according to the 1st embodiment, when a focal length of the second lens element 120 is f2, and a focal length of the third lens element 130 is f3, the following condition is satisfied:  $|f2/f3|=0.52$ .

In the optical image lens assembly according to the 1st embodiment, when the focal length of the third lens element 130 is f3, and a focal length of the fourth lens element 140 is f4, the following condition is satisfied:  $|f3/f4|=0.77$ .

In the optical image lens assembly according to the 1st embodiment, when a focal length of the first lens element 110 is f1, and the focal length of the second lens element 120 is f2, the following condition is satisfied:  $f1/f2=1.14$ .

The detailed optical data of the 1st embodiment are shown in Table 1 and the aspheric surface data are shown in Table 2 below.

TABLE 1

| 1st Embodiment                            |           |                  |           |          |       |        |              |  |
|-------------------------------------------|-----------|------------------|-----------|----------|-------|--------|--------------|--|
| f = 2.55 mm, Fno = 2.01, HFOV = 47.5 deg. |           |                  |           |          |       |        |              |  |
| Surface #                                 |           | Curvature Radius | Thickness | Material | Index | Abbe # | Focal Length |  |
| 0                                         | Object    | Plano            | Infinity  |          |       |        |              |  |
| 1                                         | Lens 1    | 1.706 ASP        | 0.410     | Plastic  | 1.544 | 55.9   | 5.63         |  |
| 2                                         |           | 3.524 ASP        | 0.115     |          |       |        |              |  |
| 3                                         | Ape. Stop | Plano            | 0.086     |          |       |        |              |  |
| 4                                         | Lens 2    | -10.627 ASP      | 0.375     | Plastic  | 1.544 | 55.9   | 4.93         |  |
| 5                                         |           | -2.167 ASP       | 0.030     |          |       |        |              |  |
| 6                                         | Lens 3    | 10.245 ASP       | 0.253     | Plastic  | 1.660 | 20.4   | -9.55        |  |
| 7                                         |           | 3.863 ASP        | 0.124     |          |       |        |              |  |
| 8                                         | Lens 4    | 5.209 ASP        | 0.310     | Plastic  | 1.544 | 55.9   | -12.36       |  |
| 9                                         |           | 2.872 ASP        | 0.177     |          |       |        |              |  |
| 10                                        | Lens 5    | -2.083 ASP       | 0.769     | Plastic  | 1.544 | 55.9   | 1.79         |  |
| 11                                        |           | -0.750 ASP       | 0.040     |          |       |        |              |  |
| 12                                        | Lens 6    | 2.671 ASP        | 0.650     | Plastic  | 1.639 | 23.5   | -2.46        |  |
| 13                                        |           | 0.895 ASP        | 0.600     |          |       |        |              |  |

TABLE 1-continued

| 1st Embodiment<br>f = 2.55 mm, Fno = 2.01, HFOV = 47.5 deg. |               |                  |           |          |       |                     |
|-------------------------------------------------------------|---------------|------------------|-----------|----------|-------|---------------------|
| Surface #                                                   |               | Curvature Radius | Thickness | Material | Index | Abbe # Focal Length |
| 14                                                          | IR-cut filter | Plano            | 0.210     | Glass    | 1.517 | 64.2 —              |
| 15                                                          |               | Plano            | 0.146     |          |       |                     |
| 16                                                          | Image         | Plano            | —         |          |       |                     |

Reference wavelength is 587.6 nm (d-line).

TABLE 2

| Aspheric Coefficients |             |             |             |             |             |             |
|-----------------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Surface #             | 1           | 2           | 4           | 5           | 6           | 7           |
| k =                   | -2.1503E-01 | -4.1194E+01 | 2.8630E+01  | -3.0290E+00 | -8.0646E+01 | -4.2570E+01 |
| A4 =                  | 1.1821E-02  | 5.2079E-02  | -1.0225E-01 | -4.3988E-01 | -3.9665E-01 | -7.9880E-02 |
| A6 =                  | -2.0540E-01 | -1.9113E-01 | -1.2111E-01 | 5.5237E-01  | 3.5070E-01  | -3.2864E-01 |
| A8 =                  | 5.7854E-01  | -5.2187E-01 | -6.3395E-01 | -8.0659E-01 | -1.0721E+00 | 8.9033E-01  |
| A10 =                 | -1.1254E+00 | 8.2953E-01  | 1.1538E+00  | -4.1537E-01 | 1.2561E+00  | -1.7592E+00 |
| A12 =                 | 6.4609E-01  |             |             | 1.7554E+00  | -1.1404E+00 | 1.8516E+00  |
| A14 =                 |             |             |             |             |             | -7.3046E-01 |
| Surface #             | 8           | 9           | 10          | 11          | 12          | 13          |
| k =                   | -2.7750E+01 | -5.2152E+01 | -2.5511E-01 | -1.3586E+00 | -8.9787E+01 | -5.3378E+00 |
| A4 =                  | -5.2751E-01 | -2.2104E-01 | -1.6243E-01 | -6.4929E-02 | 5.5725E-02  | -5.8715E-02 |
| A6 =                  | 8.5435E-01  | 5.8019E-01  | 9.8539E-01  | 3.2264E-01  | -2.3245E-01 | -2.7284E-03 |
| A8 =                  | -3.1330E+00 | -1.7138E+00 | -2.3748E+00 | -9.7532E-01 | 2.0188E-01  | 1.3158E-02  |
| A10 =                 | 5.7928E+00  | 2.7937E+00  | 4.1280E+00  | 1.2893E+00  | -7.9720E-02 | -5.8094E-03 |
| A12 =                 | -4.4041E+00 | -2.2632E+00 | -4.2626E+00 | -8.0370E-01 | 8.4974E-03  | 1.1667E-03  |
| A14 =                 | 1.1877E+00  | 6.7015E-01  | 2.1576E+00  | 2.4447E-01  | 2.1875E-03  | -1.1754E-04 |
| A16 =                 |             |             | -4.1133E-01 | -2.9550E-02 | -4.2083E-04 | 4.8616E-06  |

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In Table 1, the curvature radius, the thickness and the focal length are shown in millimeters (mm). Surface numbers 0-16 represent the surfaces sequentially arranged from the object side to the image side along the optical axis. In Table 2, k represents the conic coefficient of the equation of the aspheric surface profiles. A4-A16 represent the aspheric coefficients ranging from the 4th order to the 16th order. The tables presented below for each embodiment correspond to schematic parameter and aberration curves of each embodiment, and term definitions of the tables are the same as those in Table 1 and Table 2 of the 1st embodiment. Therefore, an explanation in this regard will not be provided again.

Furthermore, according to the 1st embodiment, when the focal length of the first lens element 110 is f1, the focal length of the second lens element 120 is f2, the focal length of the third lens element 130 is f3, the focal length of the fourth lens element 140 is f4, a focal length of the fifth lens element 150 is f5, and a focal length of the sixth lens element 160 is f6, the following conditions are satisfied:  $|f1| < |f4|$ ;  $|f2| < |f4|$ ;  $|f3| < |f4|$ ;  $|f5| < |f4|$ ; and  $|f6| < |f4|$ .

### 2nd Embodiment

FIG. 3 is a schematic view of an image capturing device according to the 2nd embodiment of the present disclosure. FIG. 4 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 2nd embodiment. In FIG. 3, the image capturing device includes an optical image lens assembly (its reference numeral is omitted) and an image sensor 290. The optical image lens assembly includes, in order from an

object side to an image side, a first lens element 210, an aperture stop 200, a second lens element 220, a third lens element 230, a fourth lens element 240, a fifth lens element 250, a sixth lens element 260, an IR-cut filter 270 and an image surface 280, wherein the image sensor 290 is disposed on the image surface 280 of the optical image lens assembly. The optical image lens assembly has a total of six lens elements (210-260), and there is an air space between every two lens elements of the first lens element 210, the second lens element 220, the third lens element 230, the fourth lens element 240, the fifth lens element 250 and the sixth lens element 260 that are adjacent to each other.

The first lens element 210 with positive refractive power has an object-side surface 211 being convex in a paraxial region thereof and an image-side surface 212 being concave in a paraxial region thereof. The first lens element 210 is made of a plastic material, and has the object-side surface 211 and the image-side surface 212 being both aspheric.

The second lens element 220 with positive refractive power has an object-side surface 221 being convex in a paraxial region thereof and an image-side surface 222 being concave in a paraxial region thereof. The second lens element 220 is made of a plastic material, and has the object-side surface 221 and the image-side surface 222 being both aspheric.

The third lens element 230 with negative refractive power has an object-side surface 231 being concave in a paraxial region thereof and an image-side surface 232 being concave in a paraxial region thereof. The third lens element 230 is made of a plastic material, and has the object-side surface 231 and the image-side surface 232 being both aspheric.

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Furthermore, the image-side surface **232** of the third lens element **230** includes at least one convex shape in an off-axial region thereof.

The fourth lens element **240** with positive refractive power has an object-side surface **241** being convex in a paraxial region thereof and an image-side surface **242** being concave in a paraxial region thereof. The fourth lens element **240** is made of a plastic material, and has the object-side surface **241** and the image-side surface **242** being both aspheric. Furthermore, the image-side surface **242** of the fourth lens element **240** includes at least one convex shape in an off-axial region thereof.

The fifth lens element **250** with positive refractive power has an object-side surface **251** being concave in a paraxial region thereof and an image-side surface **252** being convex in a paraxial region thereof. The fifth lens element **250** is made of a plastic material, and has the object-side surface **251** and the image-side surface **252** being both aspheric.

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The sixth lens element **260** with negative refractive power has an object-side surface **261** being convex in a paraxial region thereof and an image-side surface **262** being concave in a paraxial region thereof. The sixth lens element **260** is made of a plastic material, and has the object-side surface **261** and the image-side surface **262** being both aspheric. Furthermore, the image-side surface **262** of the sixth lens element **260** includes at least one convex shape in an off-axial region thereof.

The IR-cut filter **270** is made of a glass material and located between the sixth lens element **260** and the image surface **280**, and will not affect the focal length of the optical image lens assembly.

The detailed optical data of the 2nd embodiment are shown in Table 3 and the aspheric surface data are shown in Table 4 below.

TABLE 3

| 2nd Embodiment                            |               |                  |           |          |       |        |              |
|-------------------------------------------|---------------|------------------|-----------|----------|-------|--------|--------------|
| f = 2.85 mm, Fno = 2.15, HFOV = 43.7 deg. |               |                  |           |          |       |        |              |
| Surface #                                 |               | Curvature Radius | Thickness | Material | Index | Abbe # | Focal Length |
| 0                                         | Object        | Plano            | Infinity  |          |       |        |              |
| 1                                         | Lens 1        | 1.587 ASP        | 0.410     | Plastic  | 1.544 | 55.9   | 5.88         |
| 2                                         |               | 2.866 ASP        | 0.139     |          |       |        |              |
| 3                                         | Ape. Stop     | Plano            | 0.066     |          |       |        |              |
| 4                                         | Lens 2        | 100.000 ASP      | 0.341     | Plastic  | 1.544 | 55.9   | 4.89         |
| 5                                         |               | -2.729 ASP       | 0.039     |          |       |        |              |
| 6                                         | Lens 3        | -6.539 ASP       | 0.245     | Plastic  | 1.660 | 20.4   | -4.84        |
| 7                                         |               | 6.334 ASP        | 0.115     |          |       |        |              |
| 8                                         | Lens 4        | 1.646 ASP        | 0.270     | Plastic  | 1.544 | 55.9   | 12.00        |
| 9                                         |               | 2.075 ASP        | 0.374     |          |       |        |              |
| 10                                        | Lens 5        | -1.593 ASP       | 0.825     | Plastic  | 1.544 | 55.9   | 1.73         |
| 11                                        |               | -0.699 ASP       | 0.040     |          |       |        |              |
| 12                                        | Lens 6        | 2.986 ASP        | 0.596     | Plastic  | 1.639 | 23.5   | -2.00        |
| 13                                        |               | 0.825 ASP        | 0.600     |          |       |        |              |
| 14                                        | IR-cut filter | Plano            | 0.210     | Glass    | 1.517 | 64.2   | —            |
| 15                                        |               | Plano            | 0.226     |          |       |        |              |
| 16                                        | Image         | Plano            | —         |          |       |        |              |

Reference wavelength is 587.6 nm (d-line).

TABLE 4

| Aspheric Coefficients |             |             |             |             |             |             |
|-----------------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Surface #             | 1           | 2           | 4           | 5           | 6           | 7           |
| k =                   | -6.0216E-03 | -4.1194E+01 | 2.8630E+01  | -3.0290E+00 | -8.0646E+01 | -4.2570E+01 |
| A4 =                  | -6.0877E-03 | 1.4026E-01  | -1.4643E-01 | 4.0501E-02  | 1.4281E-01  | -1.7363E-01 |
| A6 =                  | -7.8322E-02 | -3.7638E-01 | -2.3012E-02 | -1.2263E+00 | -7.6316E-01 | 8.6569E-01  |
| A8 =                  | 1.8561E-01  | 1.3267E-02  | -3.2438E-01 | 2.6001E+00  | 7.1938E-02  | -2.7291E+00 |
| A10 =                 | -4.3665E-01 | 3.2456E-01  | 7.1020E-01  | -2.7647E+00 | 1.7924E+00  | 4.2459E+00  |
| A12 =                 | 2.5036E-01  |             |             | 1.1124E+00  | -2.5682E+00 | -3.6142E+00 |
| A14 =                 |             |             |             |             |             | 1.3085E+00  |
| Surface #             | 8           | 9           | 10          | 11          | 12          | 13          |
| k =                   | -2.7750E+01 | -2.5016E+01 | -5.9725E-01 | -1.3625E+00 | -8.9787E+01 | -6.5882E+00 |
| A4 =                  | 2.3826E-02  | 5.7401E-02  | -7.4731E-02 | 1.9348E-01  | 1.1273E-01  | -1.7134E-03 |
| A6 =                  | -9.2614E-01 | -4.3257E-01 | 6.2860E-01  | -2.8364E-01 | -1.6219E-01 | -2.2336E-02 |
| A8 =                  | 2.2116E+00  | 6.1584E-01  | -1.7354E+00 | 1.1670E-01  | 8.6218E-02  | 1.0000E-02  |
| A10 =                 | -3.2957E+00 | -5.2471E-01 | 2.7403E+00  | 5.0526E-02  | -2.2661E-02 | -2.0385E-03 |
| A12 =                 | 2.4462E+00  | 2.2381E-01  | -2.3844E+00 | -1.0386E-01 | 2.1360E-03  | 1.7170E-04  |
| A14 =                 | -6.5940E-01 | -3.8801E-02 | 1.0478E+00  | 6.2737E-02  | 1.6677E-04  | -9.4418E-07 |
| A16 =                 |             |             | -1.8032E-01 | -1.2502E-02 | -3.2486E-05 | -4.2277E-07 |

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In the 2nd embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 2nd embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 3 and Table 4 as the following values and satisfy the following conditions:

| 2nd Embodiment  |      |                         |       |
|-----------------|------|-------------------------|-------|
| f [mm]          | 2.85 | $(R1 + R2)/(R1 - R2)$   | -3.48 |
| Fno             | 2.15 | $(R3 + R4)/(R3 - R4)$   | 0.95  |
| HFOV [deg.]     | 43.7 | $(R9 - R10)/(R9 + R10)$ | 0.39  |
| V3/V2           | 0.36 | $f/R1 + f/R2$           | 2.79  |
| $(V3 + V6)/V5$  | 0.79 | $f2/f3$                 | 1.01  |
| CT2/T12         | 1.66 | $f3/f4$                 | 0.40  |
| Td/ $\Sigma$ AT | 4.48 | $f1/f2$                 | 1.20  |
| ImgH/f          | 1.02 |                         |       |

Furthermore, according to the 2nd embodiment, when the focal length of the first lens element **210** is  $f1$ , the focal length of the second lens element **220** is  $f2$ , the focal length of the third lens element **230** is  $f3$ , the focal length of the fourth lens element **240** is  $f4$ , a focal length of the fifth lens element **250** is  $f5$ , and a focal length of the sixth lens element **260** is  $f6$ , the following conditions are satisfied:  $|f1| < |f4|$ ;  $|f2| < |f4|$ ;  $|f3| < |f4|$ ;  $|f5| < |f4|$ ; and  $|f6| < |f4|$ .

## 3rd Embodiment

FIG. 5 is a schematic view of an image capturing device according to the 3rd embodiment of the present disclosure. FIG. 6 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 3rd embodiment. In FIG. 5, the image capturing device includes an optical image lens assembly (its reference numeral is omitted) and an image sensor **390**. The optical image lens assembly includes, in order from an object side to an image side, a first lens element **310**, an aperture stop **300**, a second lens element **320**, a third lens element **330**, a fourth lens element **340**, a fifth lens element **350**, a sixth lens element **360**, an IR-cut filter **370** and an image surface **380**, wherein the image sensor **390** is disposed on the image surface **380** of the optical image lens assembly. The optical image lens assembly has a total of six lens elements (**310-360**), and there is an air space between every two lens elements of the first lens element **310**, the second lens element **320**, the third lens element **330**, the fourth lens element **340**, the fifth lens element **350** and the sixth lens element **360** that are adjacent to each other.

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The first lens element **310** with positive refractive power has an object-side surface **311** being convex in a paraxial region thereof and an image-side surface **312** being concave in a paraxial region thereof. The first lens element **310** is made of a plastic material, and has the object-side surface **311** and the image-side surface **312** being both aspheric.

The second lens element **320** with positive refractive power has an object-side surface **321** being convex in a paraxial region thereof and an image-side surface **322** being convex in a paraxial region thereof. The second lens element **320** is made of a plastic material, and has the object-side surface **321** and the image-side surface **322** being both aspheric.

The third lens element **330** with negative refractive power has an object-side surface **331** being concave in a paraxial region thereof and an image-side surface **332** being convex in a paraxial region thereof. The third lens element **330** is made of a plastic material, and has the object-side surface **331** and the image-side surface **332** being both aspheric. Furthermore, the image-side surface **332** of the third lens element **330** includes at least one convex shape in an off-axial region thereof.

The fourth lens element **340** with positive refractive power has an object-side surface **341** being convex in a paraxial region thereof and an image-side surface **342** being concave in a paraxial region thereof. The fourth lens element **340** is made of a plastic material, and has the object-side surface **341** and the image-side surface **342** being both aspheric. Furthermore, the image-side surface **342** of the fourth lens element **340** includes at least one convex shape in an off-axial region thereof.

The fifth lens element **350** with positive refractive power has an object-side surface **351** being concave in a paraxial region thereof and an image-side surface **352** being convex in a paraxial region thereof. The fifth lens element **350** is made of a plastic material, and has the object-side surface **351** and the image-side surface **352** being both aspheric.

The sixth lens element **360** with negative refractive power has an object-side surface **361** being convex in a paraxial region thereof and an image-side surface **362** being concave in a paraxial region thereof. The sixth lens element **360** is made of a plastic material, and has the object-side surface **361** and the image-side surface **362** being both aspheric. Furthermore, the image-side surface **362** of the sixth lens element **360** includes at least one convex shape in an off-axial region thereof.

The IR-cut filter **370** is made of a glass material and located between the sixth lens element **360** and the image surface **380**, and will not affect the focal length of the optical image lens assembly.

The detailed optical data of the 3rd embodiment are shown in Table 5 and the aspheric surface data are shown in Table 6 below.

TABLE 5

| 3rd Embodiment                            |           |                  |           |          |       |        |              |
|-------------------------------------------|-----------|------------------|-----------|----------|-------|--------|--------------|
| f = 2.42 mm, Fno = 2.08, HFOV = 50.0 deg. |           |                  |           |          |       |        |              |
| Surface #                                 |           | Curvature Radius | Thickness | Material | Index | Abbe # | Focal Length |
| 0                                         | Object    | Plano            | Infinity  |          |       |        |              |
| 1                                         | Lens 1    | 1.838 ASP        | 0.410     | Plastic  | 1.583 | 30.2   | 11.40        |
| 2                                         |           | 2.330 ASP        | 0.080     |          |       |        |              |
| 3                                         | Ape. Stop | Plano            | 0.061     |          |       |        |              |
| 4                                         | Lens 2    | 19.997 ASP       | 0.489     | Plastic  | 1.544 | 55.9   | 2.49         |
| 5                                         |           | -1.440 ASP       | 0.037     |          |       |        |              |
| 6                                         | Lens 3    | -2.072 ASP       | 0.245     | Plastic  | 1.660 | 20.4   | -3.26        |

TABLE 5-continued

| 3rd Embodiment<br>f = 2.42 mm, Fno = 2.08, HFOV = 50.0 deg. |               |                  |           |          |         |        |              |
|-------------------------------------------------------------|---------------|------------------|-----------|----------|---------|--------|--------------|
| Surface #                                                   |               | Curvature Radius | Thickness | Material | Index   | Abbe # | Focal Length |
| 7                                                           |               | -59.685          | ASP       | 0.040    |         |        |              |
| 8                                                           | Lens 4        | 1.661            | ASP       | 0.280    | Plastic | 1.544  | 55.9         |
| 9                                                           |               | 1.958            | ASP       | 0.300    |         |        |              |
| 10                                                          | Lens 5        | -1.423           | ASP       | 0.911    | Plastic | 1.544  | 55.9         |
| 11                                                          |               | -0.622           | ASP       | 0.040    |         |        |              |
| 12                                                          | Lens 6        | 2.160            | ASP       | 0.513    | Plastic | 1.660  | 20.4         |
| 13                                                          |               | 0.750            | ASP       | 0.600    |         |        |              |
| 14                                                          | IR-cut filter | Plano            | 0.210     | Glass    | 1.517   | 64.2   | —            |
| 15                                                          |               | Plano            | 0.284     |          |         |        |              |
| 16                                                          | Image         | Plano            | —         |          |         |        |              |

Reference wavelength is 587.6 nm (d-line).

TABLE 6

| Aspheric Coefficients |             |             |             |             |             |             |
|-----------------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Surface #             | 1           | 2           | 4           | 5           | 6           | 7           |
| k =                   | -4.1040E-01 | -4.1194E+01 | 2.8630E+01  | -3.0290E+00 | -8.0646E+01 | -4.2570E+01 |
| A4 =                  | 1.1849E-03  | 3.3925E-01  | -1.2712E-01 | 6.0745E-01  | 3.9315E-01  | 1.8559E-01  |
| A6 =                  | -7.1697E-02 | -9.1908E-01 | 4.1987E-02  | -7.0356E+00 | -4.9663E+00 | -1.4987E+00 |
| A8 =                  | 1.6103E-01  | 7.9104E-01  | -1.2863E+00 | 2.2785E+01  | 1.4824E+01  | 3.8013E+00  |
| A10 =                 | -4.2626E-01 | 4.7133E-02  | 7.4883E-01  | -3.7683E+01 | -2.3290E+01 | -5.6249E+00 |
| A12 =                 | 3.0761E-01  |             |             | 2.4245E+01  | 1.3866E+01  | 4.2272E+00  |
| A14 =                 |             |             |             |             |             | -1.2202E+00 |
| Surface #             | 8           | 9           | 10          | 11          | 12          | 13          |
| k =                   | -2.7750E+01 | -4.6178E+01 | -1.4644E+00 | -1.3332E+00 | -8.9787E+01 | -7.0448E+00 |
| A4 =                  | -2.7542E-01 | 1.0490E-01  | -2.2235E-01 | 1.8391E-01  | 1.7665E-01  | 3.9007E-02  |
| A6 =                  | -3.5241E-01 | -6.0762E-01 | 1.1285E+00  | -3.5907E-01 | -2.1985E-01 | -6.0589E-02 |
| A8 =                  | 1.3011E-02  | 8.1641E-01  | -1.7177E+00 | 3.3578E-01  | 1.2665E-01  | 3.0340E-02  |
| A10 =                 | 1.1294E+00  | -4.6334E-01 | 1.8140E+00  | -2.3879E-01 | -4.2096E-02 | -8.3487E-03 |
| A12 =                 | -1.0260E+00 | 2.7296E-03  | -1.4254E+00 | 1.6147E-01  | 7.5311E-03  | 1.2837E-03  |
| A14 =                 | 2.6710E-01  | 5.0326E-02  | 6.3181E-01  | -6.3101E-02 | -5.8985E-04 | -1.0335E-04 |
| A16 =                 |             |             | -1.0989E-01 | 9.3837E-03  | 6.5861E-06  | 3.3717E-06  |

In the 3rd embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 3rd embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 5 and Table 6 as the following values and satisfy the following conditions:

| 3rd Embodiment |      |                       |       |
|----------------|------|-----------------------|-------|
| f [mm]         | 2.42 | (R1 + R2)/(R1 - R2)   | -8.46 |
| Fno            | 2.08 | (R3 + R4)/(R3 - R4)   | 0.87  |
| HFOV [deg.]    | 50.0 | (R9 - R10)/(R9 + R10) | 0.39  |
| V3/V2          | 0.36 | f/R1 + f/R2           | 2.36  |
| (V3 + V6)/V5   | 0.73 | f2/f3                 | 0.76  |
| CT2/T12        | 3.47 | f3/f4                 | 0.22  |
| Td/ΣAT         | 6.10 | f1/f2                 | 4.58  |
| ImgH/f         | 1.20 |                       |       |

Furthermore, according to the 3rd embodiment, when the focal length of the first lens element 310 is f1, the focal length of the second lens element 320 is f2, the focal length of the third lens element 330 is f3, the focal length of the

fourth lens element 340 is f4, a focal length of the fifth lens element 350 is f5, and a focal length of the sixth lens element 360 is f6, the following conditions are satisfied:  $|f1| < |f4|$ ;  $|f2| < |f4|$ ;  $|f3| < |f4|$ ;  $|f5| < |f4|$ ; and  $|f6| < |f4|$ .

#### 4th Embodiment

FIG. 7 is a schematic view of an image capturing device according to the 4th embodiment of the present disclosure. FIG. 8 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 4th embodiment. In FIG. 7, the image capturing device includes an optical image lens assembly (its reference numeral is omitted) and an image sensor 490. The optical image lens assembly includes, in order from an object side to an image side, a first lens element 410, an aperture stop 400, a second lens element 420, a third lens element 430, a fourth lens element 440, a fifth lens element 450, a sixth lens element 460, an IR-cut filter 470 and an image surface 480, wherein the image sensor 490 is disposed on the image surface 480 of the optical image lens assembly. The optical image lens assembly has a total of six lens elements (410-460), and there is an air space between every two lens elements of the first lens element 410, the second lens element 420, the third lens element 430, the fourth lens element 440, the fifth lens element 450 and the sixth lens element 460 that are adjacent to each other.

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The first lens element **410** with positive refractive power has an object-side surface **411** being convex in a paraxial region thereof and an image-side surface **412** being concave in a paraxial region thereof. The first lens element **410** is made of a plastic material, and has the object-side surface **411** and the image-side surface **412** being both aspheric.

The second lens element **420** with positive refractive power has an object-side surface **421** being concave in a paraxial region thereof and an image-side surface **422** being convex in a paraxial region thereof. The second lens element **420** is made of a plastic material, and has the object-side surface **421** and the image-side surface **422** being both aspheric.

The third lens element **430** with negative refractive power has an object-side surface **431** being concave in a paraxial region thereof and an image-side surface **432** being concave in a paraxial region thereof. The third lens element **430** is made of a plastic material, and has the object-side surface **431** and the image-side surface **432** being both aspheric. Furthermore, the image-side surface **432** of the third lens element **430** includes at least one convex shape in an off-axial region thereof.

The fourth lens element **440** with negative refractive power has an object-side surface **441** being convex in a paraxial region thereof and an image-side surface **442** being concave in a paraxial region thereof. The fourth lens element

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**440** is made of a plastic material, and has the object-side surface **441** and the image-side surface **442** being both aspheric. Furthermore, the image-side surface **442** of the fourth lens element **440** includes at least one convex shape in an off-axial region thereof.

The fifth lens element **450** with positive refractive power has an object-side surface **451** being concave in a paraxial region thereof and an image-side surface **452** being convex in a paraxial region thereof. The fifth lens element **450** is made of a plastic material, and has the object-side surface **451** and the image-side surface **452** being both aspheric.

The sixth lens element **460** with negative refractive power has an object-side surface **461** being convex in a paraxial region thereof and an image-side surface **462** being concave in a paraxial region thereof. The sixth lens element **460** is made of a plastic material, and has the object-side surface **461** and the image-side surface **462** being both aspheric. Furthermore, the image-side surface **462** of the sixth lens element **460** includes at least one convex shape in an off-axial region thereof.

The IR-cut filter **470** is made of a glass material and located between the sixth lens element **460** and the image surface **480**, and will not affect the focal length of the optical image lens assembly.

The detailed optical data of the 4th embodiment are shown in Table 7 and the aspheric surface data are shown in Table 8 below.

TABLE 7

| 4th Embodiment                            |               |                  |           |          |       |        |              |
|-------------------------------------------|---------------|------------------|-----------|----------|-------|--------|--------------|
| f = 2.54 mm, Fno = 1.89, HFOV = 47.5 deg. |               |                  |           |          |       |        |              |
| Surface #                                 |               | Curvature Radius | Thickness | Material | Index | Abbe # | Focal Length |
| 0                                         | Object        | Plano            | Infinity  |          |       |        |              |
| 1                                         | Lens 1        | 1.571 ASP        | 0.443     | Plastic  | 1.544 | 55.9   | 4.69         |
| 2                                         |               | 3.684 ASP        | 0.072     |          |       |        |              |
| 3                                         | Ape. Stop     | Plano            | 0.105     |          |       |        |              |
| 4                                         | Lens 2        | -7.093 ASP       | 0.396     | Plastic  | 1.544 | 55.9   | 5.53         |
| 5                                         |               | -2.153 ASP       | 0.031     |          |       |        |              |
| 6                                         | Lens 3        | -64.832 ASP      | 0.250     | Plastic  | 1.660 | 20.4   | -6.81        |
| 7                                         |               | 4.833 ASP        | 0.113     |          |       |        |              |
| 8                                         | Lens 4        | 6.152 ASP        | 0.280     | Plastic  | 1.544 | 55.9   | -21.15       |
| 9                                         |               | 3.944 ASP        | 0.146     |          |       |        |              |
| 10                                        | Lens 5        | -2.777 ASP       | 0.880     | Plastic  | 1.544 | 55.9   | 1.27         |
| 11                                        |               | -0.616 ASP       | 0.045     |          |       |        |              |
| 12                                        | Lens 6        | 1.796 ASP        | 0.390     | Plastic  | 1.576 | 36.4   | -1.47        |
| 13                                        |               | 0.530 ASP        | 0.600     |          |       |        |              |
| 14                                        | IR-cut filter | Plano            | 0.210     | Glass    | 1.517 | 64.2   | —            |
| 15                                        |               | Plano            | 0.226     |          |       |        |              |
| 16                                        | Image         | Plano            | —         |          |       |        |              |

Reference wavelength is 587.6 nm (d-line).

Effective radius of Surface 11 is 1.375 mm.

TABLE 8

| Aspheric Coefficients |             |             |             |             |             |             |
|-----------------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Surface #             | 1           | 2           | 4           | 5           | 6           | 7           |
| k =                   | 3.8806E-01  | -2.0686E+01 | -1.1894E+01 | -3.8728E+00 | -9.0000E+01 | -6.9165E+01 |
| A4 =                  | 3.1918E-03  | 3.8720E-02  | -7.3617E-02 | -5.6101E-01 | -6.0897E-01 | -2.2355E-01 |
| A6 =                  | -3.4770E-02 | 1.5007E-02  | -1.4305E-01 | 1.6339E+00  | 1.5380E+00  | 1.9106E-01  |
| A8 =                  | 9.3345E-02  | -5.1553E-01 | -1.4626E-01 | -4.9427E+00 | -4.9250E+00 | -3.2132E-01 |
| A10 =                 | -1.6165E-01 | 4.4129E-01  | -2.0944E-01 | 6.8277E+00  | 6.9683E+00  | 5.2298E-02  |
| A12 =                 |             |             |             | -4.0008E+00 | -4.8707E+00 | 1.8367E-01  |
| A14 =                 |             |             |             |             |             | -7.1001E-02 |

TABLE 8-continued

| Aspheric Coefficients |             |             |             |             |             |             |
|-----------------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Surface #             | 8           | 9           | 10          | 11          | 12          | 13          |
| k =                   | -3.8939E+01 | -2.4429E+01 | -5.3677E-02 | -1.6632E+00 | -6.9037E+01 | -4.8536E+00 |
| A4 =                  | -4.1874E-01 | -2.3101E-01 | -6.5951E-03 | 3.2129E-01  | 3.4875E-02  | -6.0023E-02 |
| A6 =                  | 1.3896E-01  | 1.3286E-01  | 2.4241E-02  | -9.7452E-01 | -2.3076E-01 | -2.2081E-03 |
| A8 =                  | -2.8309E-01 | -2.1920E-01 | 1.5483E-01  | 1.3826E+00  | 1.9065E-01  | 1.2101E-02  |
| A10 =                 | 5.5587E-01  | 1.2656E-01  | -6.2336E-01 | -1.2815E+00 | -7.6828E-02 | -5.7736E-03 |
| A12 =                 | -2.3905E-01 | -8.3785E-02 | 6.3957E-01  | 7.4642E-01  | 1.1276E-02  | 1.2473E-03  |
| A14 =                 | 1.0260E-02  | 4.9684E-02  | -2.3663E-01 | -2.2554E-01 | 1.1160E-03  | -1.3333E-04 |
| A16 =                 |             |             | 2.4819E-02  | 2.6122E-02  | -3.3069E-04 | 5.7747E-06  |

In the 4th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 4th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 7 and Table 8 as the following values and satisfy the following conditions:

| 4th Embodiment  |      |                         |       |
|-----------------|------|-------------------------|-------|
| f [mm]          | 2.54 | $(R1 + R2)/(R1 - R2)$   | -2.49 |
| Fno             | 1.89 | $(R3 + R4)/(R3 - R4)$   | 1.87  |
| HFOV [deg.]     | 47.5 | $(R9 - R10)/(R9 + R10)$ | 0.64  |
| V3/V2           | 0.36 | $f/R1 + f/R2$           | 2.31  |
| $(V3 + V6)/V5$  | 1.02 | $ f2/f3 $               | 0.81  |
| CT2/T12         | 2.24 | $ f3/f4 $               | 0.32  |
| Td/ $\Sigma$ AT | 6.15 | $f1/f2$                 | 0.85  |
| ImgH/f          | 1.14 |                         |       |

Furthermore, according to the 4th embodiment, when the focal length of the first lens element **410** is  $f1$ , the focal length of the second lens element **420** is  $f2$ , the focal length of the third lens element **430** is  $f3$ , the focal length of the fourth lens element **440** is  $f4$ , a focal length of the fifth lens element **450** is  $f5$ , and a focal length of the sixth lens element **460** is  $f6$ , the following conditions are satisfied:  $|f1| < |f4|$ ;  $|f2| < |f4|$ ;  $|f3| < |f4|$ ;  $|f5| < |f4|$ ; and  $|f6| < |f4|$ .

#### 5th Embodiment

FIG. 9 is a schematic view of an image capturing device according to the 5th embodiment of the present disclosure. FIG. 10 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 5th embodiment. In FIG. 9, the image capturing device includes an optical image lens assembly (its reference numeral is omitted) and an image sensor **590**. The optical image lens assembly includes, in order from an object side to an image side, a first lens element **510**, an aperture stop **500**, a second lens element **520**, a third lens element **530**, a fourth lens element **540**, a fifth lens element **550**, a sixth lens element **560**, an IR-cut filter **570** and an image surface **580**, wherein the image sensor **590** is disposed on the image surface **580** of the optical image lens assembly. The optical image lens assembly has a total of six lens elements (**510-560**), and there is an air space between every two lens elements of the first lens element **510**, the second lens element **520**, the third lens element **530**, the fourth lens element **540**, the fifth lens element **550** and the sixth lens element **560** that are adjacent to each other.

The first lens element **510** with positive refractive power has an object-side surface **511** being convex in a paraxial region thereof and an image-side surface **512** being concave in a paraxial region thereof. The first lens element **510** is made of a plastic material, and has the object-side surface **511** and the image-side surface **512** being both aspheric.

The second lens element **520** with positive refractive power has an object-side surface **521** being concave in a paraxial region thereof and an image-side surface **522** being convex in a paraxial region thereof. The second lens element **520** is made of a plastic material, and has the object-side surface **521** and the image-side surface **522** being both aspheric.

The third lens element **530** with negative refractive power has an object-side surface **531** being convex in a paraxial region thereof and an image-side surface **532** being concave in a paraxial region thereof. The third lens element **530** is made of a plastic material, and has the object-side surface **531** and the image-side surface **532** being both aspheric. Furthermore, the image-side surface **532** of the third lens element **530** includes at least one convex shape in an off-axial region thereof.

The fourth lens element **540** with positive refractive power has an object-side surface **541** being convex in a paraxial region thereof and an image-side surface **542** being concave in a paraxial region thereof. The fourth lens element **540** is made of a plastic material, and has the object-side surface **541** and the image-side surface **542** being both aspheric. Furthermore, the image-side surface **542** of the fourth lens element **540** includes at least one convex shape in an off-axial region thereof.

The fifth lens element **550** with positive refractive power has an object-side surface **551** being concave in a paraxial region thereof and an image-side surface **552** being convex in a paraxial region thereof. The fifth lens element **550** is made of a plastic material, and has the object-side surface **551** and the image-side surface **552** being both aspheric.

The sixth lens element **560** with negative refractive power has an object-side surface **561** being convex in a paraxial region thereof and an image-side surface **562** being concave in a paraxial region thereof. The sixth lens element **560** is made of a plastic material, and has the object-side surface **561** and the image-side surface **562** being both aspheric. Furthermore, the image-side surface **562** of the sixth lens element **560** includes at least one convex shape in an off-axial region thereof.

The IR-cut filter **570** is made of a glass material and located between the sixth lens element **560** and the image surface **580**, and will not affect the focal length of the optical image lens assembly.

The detailed optical data of the 5th embodiment are shown in Table 9 and the aspheric surface data are shown in Table 10 below.

TABLE 9

| 5th Embodiment                            |               |                  |           |          |       |        |              |  |
|-------------------------------------------|---------------|------------------|-----------|----------|-------|--------|--------------|--|
| f = 2.54 mm, Fno = 1.89, HFOV = 47.5 deg. |               |                  |           |          |       |        |              |  |
| Surface #                                 |               | Curvature Radius | Thickness | Material | Index | Abbe # | Focal Length |  |
| 0                                         | Object        | Plano            | Infinity  |          |       |        |              |  |
| 1                                         | Lens 1        | 1.750 ASP        | 0.412     | Plastic  | 1.544 | 55.9   | 5.08         |  |
| 2                                         |               | 4.384 ASP        | 0.066     |          |       |        |              |  |
| 3                                         | Ape. Stop     | Plano            | 0.108     |          |       |        |              |  |
| 4                                         | Lens 2        | -7.437 ASP       | 0.362     | Plastic  | 1.544 | 55.9   | 6.64         |  |
| 5                                         |               | -2.471 ASP       | 0.030     |          |       |        |              |  |
| 6                                         | Lens 3        | 22.963 ASP       | 0.245     | Plastic  | 1.660 | 20.4   | -7.79        |  |
| 7                                         |               | 4.181 ASP        | 0.115     |          |       |        |              |  |
| 8                                         | Lens 4        | 2.698 ASP        | 0.270     | Plastic  | 1.544 | 55.9   | 866.84       |  |
| 9                                         |               | 2.618 ASP        | 0.216     |          |       |        |              |  |
| 10                                        | Lens 5        | -2.282 ASP       | 0.811     | Plastic  | 1.544 | 55.9   | 1.50         |  |
| 11                                        |               | -0.675 ASP       | 0.040     |          |       |        |              |  |
| 12                                        | Lens 6        | 2.264 ASP        | 0.500     | Plastic  | 1.583 | 30.2   | -1.78        |  |
| 13                                        |               | 0.653 ASP        | 0.600     |          |       |        |              |  |
| 14                                        | IR-cut filter | Plano            | 0.210     | Glass    | 1.517 | 64.2   | —            |  |
| 15                                        |               | Plano            | 0.205     |          |       |        |              |  |
| 16                                        | Image         | Plano            | —         |          |       |        |              |  |

Reference wavelength is 587.6 nm (d-line).

Effective radius of Surface 11 is 1.320 mm.

TABLE 10

| Aspheric Coefficients |             |             |             |             |             |             |
|-----------------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Surface #             | 1           | 2           | 4           | 5           | 6           | 7           |
| k =                   | 3.5090E-01  | -4.1205E+01 | 2.8589E+01  | -3.0288E+00 | -8.0646E+01 | -4.2603E+01 |
| A4 =                  | 2.0407E-03  | 9.7330E-03  | -8.4349E-02 | -4.7480E-01 | -4.7314E-01 | -2.4028E-01 |
| A6 =                  | -1.5311E-01 | -4.5218E-02 | -2.1297E-01 | 9.7456E-01  | 1.1754E+00  | 5.4001E-01  |
| A8 =                  | 4.4213E-01  | -5.3666E-01 | 1.6985E-01  | -3.0807E+00 | -4.4916E+00 | -1.4776E+00 |
| A10 =                 | -8.3069E-01 | 7.3645E-01  | -2.5003E-01 | 5.0900E+00  | 7.7637E+00  | 2.0439E+00  |
| A12 =                 | 4.6457E-01  |             |             | -3.8918E+00 | -6.3890E+00 | -1.7119E+00 |
| A14 =                 |             |             |             |             |             | 6.6847E-01  |
| Surface #             | 8           | 9           | 10          | 11          | 12          | 13          |
| k =                   | -2.7726E+01 | -1.3062E+01 | -1.9707E+00 | -1.4588E+00 | -8.9802E+01 | -5.0501E+00 |
| A4 =                  | -4.2124E-01 | -2.1776E-01 | 9.9938E-04  | 1.5454E-01  | 4.2992E-02  | -6.3516E-02 |
| A6 =                  | 1.5364E-01  | 2.4565E-02  | 1.9611E-01  | -2.1761E-01 | -2.5011E-01 | 6.2235E-03  |
| A8 =                  | -9.8882E-02 | 8.7339E-03  | -7.5322E-01 | -2.1552E-01 | 2.3160E-01  | 6.8251E-03  |
| A10 =                 | -5.7366E-02 | 6.7631E-02  | 1.6383E+00  | 6.7642E-01  | -1.0894E-01 | -3.9389E-03 |
| A12 =                 | 3.8948E-01  | -2.1896E-01 | -2.0165E+00 | -6.3293E-01 | 2.3487E-02  | 9.0671E-04  |
| A14 =                 | -1.9578E-01 | 1.2259E-01  | 1.2090E+00  | 2.9289E-01  | -1.1284E-03 | -1.0116E-04 |
| A16 =                 |             |             | -2.7159E-01 | -5.3969E-02 | -1.7368E-04 | 4.5113E-06  |

In the 5th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 5th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 9 and Table 10 as the following values and satisfy the following conditions:

| 5th Embodiment |      |                       |       |
|----------------|------|-----------------------|-------|
| f [mm]         | 2.54 | (R1 + R2)/(R1 - R2)   | -2.33 |
| Fno            | 1.89 | (R3 + R4)/(R3 - R4)   | 2.00  |
| HFOV [deg.]    | 47.5 | (R9 - R10)/(R9 + R10) | 0.54  |
| V3/V2          | 0.36 | f/R1 + f/R2           | 2.03  |
| (V3 + V6)/V5   | 0.91 | f2/f3                 | 0.85  |
| CT2/T12        | 2.08 | f3/f4                 | 0.01  |

-continued

| 5th Embodiment |      |       |      |
|----------------|------|-------|------|
| Td/ΣAT         | 5.52 | f1/f2 | 0.77 |
| ImgH/f         | 1.14 |       |      |

Furthermore, according to the 5th embodiment, when the focal length of the first lens element **510** is f1, the focal length of the second lens element **520** is f2, the focal length of the third lens element **530** is f3, the focal length of the fourth lens element **540** is f4, a focal length of the fifth lens element **550** is f5, and a focal length of the sixth lens element **560** is f6, the following conditions are satisfied: |f1|<|f4|; |f2|<|f4|; |f3|<|f4|; |f5|<|f4|; and |f6|<|f4|.

6th Embodiment

FIG. 11 is a schematic view of an image capturing device according to the 6th embodiment of the present disclosure.

FIG. 12 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 6th embodiment. In FIG. 11, the image capturing device includes an optical image lens assembly (its reference numeral is omitted) and an image sensor 690. The optical image lens assembly includes, in order from an object side to an image side, an aperture stop 600, a first lens element 610, a second lens element 620, a third lens element 630, a fourth lens element 640, a fifth lens element 650, a sixth lens element 660, an IR-cut filter 670 and an image surface 680, wherein the image sensor 690 is disposed on the image surface 680 of the optical image lens assembly. The optical image lens assembly has a total of six lens elements (610-660), and there is an air space between every two lens elements of the first lens element 610, the second lens element 620, the third lens element 630, the fourth lens element 640, the fifth lens element 650 and the sixth lens element 660 that are adjacent to each other.

The first lens element 610 with positive refractive power has an object-side surface 611 being convex in a paraxial region thereof and an image-side surface 612 being concave in a paraxial region thereof. The first lens element 610 is made of a plastic material, and has the object-side surface 611 and the image-side surface 612 being both aspheric.

The second lens element 620 with positive refractive power has an object-side surface 621 being convex in a paraxial region thereof and an image-side surface 622 being concave in a paraxial region thereof. The second lens element 620 is made of a plastic material, and has the object-side surface 621 and the image-side surface 622 being both aspheric.

The third lens element 630 with negative refractive power has an object-side surface 631 being convex in a paraxial region thereof and an image-side surface 632 being concave

in a paraxial region thereof. The third lens element 630 is made of a plastic material, and has the object-side surface 631 and the image-side surface 632 being both aspheric.

The fourth lens element 640 with positive refractive power has an object-side surface 641 being convex in a paraxial region thereof and an image-side surface 642 being concave in a paraxial region thereof. The fourth lens element 640 is made of a plastic material, and has the object-side surface 641 and the image-side surface 642 being both aspheric. Furthermore, the image-side surface 642 of the fourth lens element 640 includes at least one convex shape in an off-axial region thereof.

The fifth lens element 650 with positive refractive power has an object-side surface 651 being concave in a paraxial region thereof and an image-side surface 652 being convex in a paraxial region thereof. The fifth lens element 650 is made of a plastic material, and has the object-side surface 651 and the image-side surface 652 being both aspheric.

The sixth lens element 660 with negative refractive power has an object-side surface 661 being concave in a paraxial region thereof and an image-side surface 662 being concave in a paraxial region thereof. The sixth lens element 660 is made of a plastic material, and has the object-side surface 661 and the image-side surface 662 being both aspheric. Furthermore, the image-side surface 662 of the sixth lens element 660 includes at least one convex shape in an off-axial region thereof.

The IR-cut filter 670 is made of a glass material and located between the sixth lens element 660 and the image surface 680, and will not affect the focal length of the optical image lens assembly.

The detailed optical data of the 6th embodiment are shown in Table 11 and the aspheric surface data are shown in Table 12 below.

TABLE 11

| 6th Embodiment                            |               |                  |           |          |       |        |              |  |
|-------------------------------------------|---------------|------------------|-----------|----------|-------|--------|--------------|--|
| f = 3.64 mm, Fno = 2.08, HFOV = 38.3 deg. |               |                  |           |          |       |        |              |  |
| Surface #                                 |               | Curvature Radius | Thickness | Material | Index | Abbe # | Focal Length |  |
| 0                                         | Object        | Plano            | Infinity  |          |       |        |              |  |
| 1                                         | Ape. Stop     | Plano            | -0.140    |          |       |        |              |  |
| 2                                         | Lens 1        | 1.993 ASP        | 0.400     | Plastic  | 1.544 | 55.9   | 4.91         |  |
| 3                                         |               | 7.299 ASP        | 0.080     |          |       |        |              |  |
| 4                                         | Lens 2        | 12.573 ASP       | 0.387     | Plastic  | 1.544 | 55.9   | 5.48         |  |
| 5                                         |               | -3.862 ASP       | 0.050     |          |       |        |              |  |
| 6                                         | Lens 3        | 5.844 ASP        | 0.240     | Plastic  | 1.660 | 20.4   | -4.88        |  |
| 7                                         |               | 2.041 ASP        | 0.442     |          |       |        |              |  |
| 8                                         | Lens 4        | 34.452 ASP       | 0.346     | Plastic  | 1.583 | 30.2   | 96.16        |  |
| 9                                         |               | 88.957 ASP       | 0.404     |          |       |        |              |  |
| 10                                        | Lens 5        | -96.971 ASP      | 0.537     | Plastic  | 1.544 | 55.9   | 2.35         |  |
| 11                                        |               | -1.264 ASP       | 0.161     |          |       |        |              |  |
| 12                                        | Lens 6        | -19.641 ASP      | 0.580     | Plastic  | 1.544 | 55.9   | -1.82        |  |
| 13                                        |               | 1.051 ASP        | 0.400     |          |       |        |              |  |
| 14                                        | IR-cut filter | Plano            | 0.210     | Glass    | 1.517 | 64.2   | —            |  |
| 15                                        |               | Plano            | 0.362     |          |       |        |              |  |
| 16                                        | Image         | Plano            | —         |          |       |        |              |  |

Reference wavelength is 587.6 nm (d-line).

Effective radius of Surface 6 is 0.900 mm.

TABLE 12

| Aspheric Coefficients |             |             |             |             |             |             |
|-----------------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Surface #             | 2           | 3           | 4           | 5           | 6           | 7           |
| k =                   | -1.7676E+00 | 4.1365E+01  | -4.0466E+01 | -6.4965E+01 | 4.4826E+00  | 1.2843E+00  |
| A4 =                  | -2.3303E-02 | -1.4152E-01 | -7.7861E-02 | -6.9569E-02 | -3.9605E-02 | -1.3279E-01 |

TABLE 12-continued

| Aspheric Coefficients |             |             |             |             |             |             |
|-----------------------|-------------|-------------|-------------|-------------|-------------|-------------|
| A6 =                  | -2.1029E-02 | -2.1231E-02 | 3.4944E-02  | -3.5328E-02 | -1.2894E-01 | 1.2000E-01  |
| A8 =                  | -7.2917E-02 | 1.5744E-01  | 1.9389E-01  | 2.3263E-01  | 2.2839E-01  | -2.1210E-01 |
| A10 =                 | 4.7624E-02  | -1.9157E-01 | -1.3714E-01 | -5.0542E-01 | -2.6515E-01 | 2.9793E-01  |
| A12 =                 | -4.7687E-03 | 3.0087E-01  | 1.4510E-01  | 5.5354E-01  | 2.4064E-01  | -2.0723E-01 |
| A14 =                 | 4.0362E-03  | -2.0278E-01 | -1.3147E-01 | -2.4425E-01 | -9.7955E-02 | 6.1362E-02  |
| Surface #             | 8           | 9           | 10          | 11          | 12          | 13          |
| k =                   | 2.0000E-01  | 2.0000E+01  | -1.0001E+00 | -6.3148E+00 | 9.0000E+01  | -6.1361E+00 |
| A4 =                  | -1.1437E-01 | -9.3368E-02 | 1.5437E-01  | 9.2905E-02  | -1.1410E-01 | -9.3815E-02 |
| A6 =                  | -1.4765E-01 | -1.8200E-01 | -2.1261E-01 | -9.1931E-02 | -7.5689E-02 | 4.0481E-02  |
| A8 =                  | 4.2302E-01  | 2.5228E-01  | 1.1980E-01  | 5.0231E-03  | 7.5124E-02  | -1.3111E-02 |
| A10 =                 | -6.3705E-01 | -1.6790E-01 | -4.7933E-02 | 2.1835E-02  | -1.3047E-02 | 3.0547E-03  |
| A12 =                 | 4.5648E-01  | 1.9713E-02  | 1.0965E-02  | -8.7235E-03 | -2.9724E-03 | -5.0462E-04 |
| A14 =                 | -1.2291E-01 | 3.0755E-02  | -5.0670E-04 | 1.2469E-03  | 1.1954E-03  | 5.0175E-05  |
| A16 =                 |             | -8.4157E-03 | -1.2874E-04 | -6.0136E-05 | -1.0703E-04 | -2.1647E-06 |

In the 6th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 6th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 11 and Table 12 as the following values and satisfy the following conditions:

| 6th Embodiment  |      |                         |       |
|-----------------|------|-------------------------|-------|
| f [mm]          | 3.64 | $(R1 + R2)/(R1 - R2)$   | -1.75 |
| Fno             | 2.08 | $(R3 + R4)/(R3 - R4)$   | 0.53  |
| HFOV [deg.]     | 38.3 | $(R9 - R10)/(R9 + R10)$ | 0.97  |
| V3/V2           | 0.36 | $f/R1 + f/R2$           | 2.33  |
| $(V3 + V6)/V5$  | 1.36 | $f2/f3$                 | 1.12  |
| CT2/T12         | 4.84 | $f3/f4$                 | 0.05  |
| Td/ $\Sigma$ AT | 3.19 | $f1/f2$                 | 0.90  |
| ImgH/f          | 0.81 |                         |       |

Furthermore, according to the 6th embodiment, when the focal length of the first lens element 610 is  $f1$ , the focal length of the second lens element 620 is  $f2$ , the focal length of the third lens element 630 is  $f3$ , the focal length of the fourth lens element 640 is  $f4$ , a focal length of the fifth lens element 650 is  $f5$ , and a focal length of the sixth lens element 660 is  $f6$ , the following conditions are satisfied:  $|f1| < |f4|$ ;  $|f2| < |f4|$ ;  $|f3| < |f4|$ ;  $|f5| < |f4|$ ; and  $|f6| < |f4|$ .

#### 7th Embodiment

FIG. 13 is a schematic view of an image capturing device according to the 7th embodiment of the present disclosure. FIG. 14 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 7th embodiment. In FIG. 13, the image capturing device includes an optical image lens assembly (its reference numeral is omitted) and an image sensor 790. The optical image lens assembly includes, in order from an object side to an image side, an aperture stop 700, a first lens element 710, a second lens element 720, a third lens element 730, a fourth lens element 740, a fifth lens element 750, a sixth lens element 760, an IR-cut filter 770 and an image surface 780, wherein the image sensor 790 is disposed on the image surface 780 of the optical image lens assembly. The optical image lens assembly has a total of six lens elements (710-760), and there is an air space between every two lens

elements of the first lens element 710, the second lens element 720, the third lens element 730, the fourth lens element 740, the fifth lens element 750 and the sixth lens element 760 that are adjacent to each other.

The first lens element 710 with positive refractive power has an object-side surface 711 being convex in a paraxial region thereof and an image-side surface 712 being concave in a paraxial region thereof. The first lens element 710 is made of a plastic material, and has the object-side surface 711 and the image-side surface 712 being both aspheric.

The second lens element 720 with positive refractive power has an object-side surface 721 being convex in a paraxial region thereof and an image-side surface 722 being convex in a paraxial region thereof. The second lens element 720 is made of a plastic material, and has the object-side surface 721 and the image-side surface 722 being both aspheric.

The third lens element 730 with negative refractive power has an object-side surface 731 being convex in a paraxial region thereof and an image-side surface 732 being concave in a paraxial region thereof. The third lens element 730 is made of a plastic material, and has the object-side surface 731 and the image-side surface 732 being both aspheric.

The fourth lens element 740 with positive refractive power has an object-side surface 741 being convex in a paraxial region thereof and an image-side surface 742 being concave in a paraxial region thereof. The fourth lens element 740 is made of a plastic material, and has the object-side surface 741 and the image-side surface 742 being both aspheric. Furthermore, the image-side surface 742 of the fourth lens element 740 includes at least one convex shape in an off-axial region thereof.

The fifth lens element 750 with positive refractive power has an object-side surface 751 being concave in a paraxial region thereof and an image-side surface 752 being convex in a paraxial region thereof. The fifth lens element 750 is made of a plastic material, and has the object-side surface 751 and the image-side surface 752 being both aspheric.

The sixth lens element 760 with negative refractive power has an object-side surface 761 being convex in a paraxial region thereof and an image-side surface 762 being concave in a paraxial region thereof. The sixth lens element 760 is made of a plastic material, and has the object-side surface 761 and the image-side surface 762 being both aspheric. Furthermore, the image-side surface 762 of the sixth lens element 760 includes at least one convex shape in an off-axial region thereof.

The IR-cut filter **770** is made of a glass material and located between the sixth lens element **760** and the image surface **780**, and will not affect the focal length of the optical image lens assembly.

The detailed optical data of the 7th embodiment are shown in Table 13 and the aspheric surface data are shown in Table 14 below.

TABLE 13

| 7th Embodiment                            |               |                  |           |          |       |        |              |  |
|-------------------------------------------|---------------|------------------|-----------|----------|-------|--------|--------------|--|
| f = 3.43 mm, Fno = 2.22, HFOV = 40.0 deg. |               |                  |           |          |       |        |              |  |
| Surface #                                 |               | Curvature Radius | Thickness | Material | Index | Abbe # | Focal Length |  |
| 0                                         | Object        | Plano            | Infinity  |          |       |        |              |  |
| 1                                         | Ape. Stop     | Plano            | -0.069    |          |       |        |              |  |
| 2                                         | Lens 1        | 2.549 ASP        | 0.400     | Plastic  | 1.544 | 55.9   | 5.46         |  |
| 3                                         |               | 16.883 ASP       | 0.030     |          |       |        |              |  |
| 4                                         | Lens 2        | 15.699 ASP       | 0.506     | Plastic  | 1.544 | 55.9   | 5.05         |  |
| 5                                         |               | -3.294 ASP       | 0.050     |          |       |        |              |  |
| 6                                         | Lens 3        | 3.759 ASP        | 0.240     | Plastic  | 1.660 | 20.4   | -5.05        |  |
| 7                                         |               | 1.721 ASP        | 0.388     |          |       |        |              |  |
| 8                                         | Lens 4        | 7.160 ASP        | 0.313     | Plastic  | 1.583 | 30.2   | 104.29       |  |
| 9                                         |               | 7.984 ASP        | 0.338     |          |       |        |              |  |
| 10                                        | Lens 5        | -76.201 ASP      | 0.636     | Plastic  | 1.544 | 55.9   | 2.77         |  |
| 11                                        |               | -1.481 ASP       | 0.123     |          |       |        |              |  |
| 12                                        | Lens 6        | 4.955 ASP        | 0.620     | Plastic  | 1.544 | 55.9   | -2.24        |  |
| 13                                        |               | 0.935 ASP        | 0.400     |          |       |        |              |  |
| 14                                        | IR-cut filter | Plano            | 0.210     | Glass    | 1.517 | 64.2   | —            |  |
| 15                                        |               | Plano            | 0.338     |          |       |        |              |  |
| 16                                        | Image         | Plano            | —         |          |       |        |              |  |

Reference wavelength is 587.6 nm (d-line).

TABLE 14

| Aspheric Coefficients |             |             |             |             |             |             |
|-----------------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Surface #             | 2           | 3           | 4           | 5           | 6           | 7           |
| k =                   | -5.6336E+00 | -5.0000E+01 | -4.0466E+01 | -6.4950E+01 | -1.4713E+01 | 3.4805E-01  |
| A4 =                  | -4.4279E-02 | -3.3447E-01 | -2.3962E-01 | -2.3449E-01 | -1.5096E-01 | -2.1200E-01 |
| A6 =                  | -4.1180E-02 | 4.6442E-01  | 5.0294E-01  | 7.0020E-01  | 3.6813E-01  | 2.9980E-01  |
| A8 =                  | -9.8324E-02 | -3.0521E-02 | 1.7820E-01  | -1.6415E+00 | -1.0291E+00 | -5.5722E-01 |
| A10 =                 | 1.2782E-01  | -3.0820E-01 | -8.5788E-01 | 2.2749E+00  | 1.5219E+00  | 6.6474E-01  |
| A12 =                 | -4.3430E-03 | 5.6720E-02  | 6.0241E-01  | -1.6318E+00 | -1.0734E+00 | -4.0160E-01 |
| A14 =                 | -3.5159E-02 | 9.2771E-02  | -1.2805E-01 | 4.5587E-01  | 2.8641E-01  | 9.5013E-02  |
| Surface #             | 8           | 9           | 10          | 11          | 12          | 13          |
| k =                   | -4.0189E-01 | 5.1358E+00  | -1.0001E+00 | -2.2364E+00 | -9.0000E+01 | -4.9200E+00 |
| A4 =                  | -8.5317E-02 | -4.8393E-02 | 1.9451E-01  | 2.2585E-01  | -1.8724E-01 | -1.0117E-01 |
| A6 =                  | -1.3189E-01 | -2.5843E-01 | -2.6108E-01 | -2.1238E-01 | 3.6746E-02  | 4.3310E-02  |
| A8 =                  | 4.3396E-01  | 3.7979E-01  | 1.1326E-01  | 4.7545E-02  | -9.5437E-02 | -1.4819E-02 |
| A10 =                 | -5.4457E-01 | -2.4547E-01 | 9.6863E-03  | 2.3321E-02  | 1.0472E-01  | 3.6450E-03  |
| A12 =                 | 3.2042E-01  | 3.9778E-02  | -3.4904E-02 | -1.3014E-02 | -4.2850E-02 | -5.8560E-04 |
| A14 =                 | -7.0672E-02 | 2.6315E-02  | 1.3453E-02  | 2.1489E-03  | 7.8329E-03  | 5.2083E-05  |
| A16 =                 |             | -7.6246E-03 | -1.5651E-03 | -1.1816E-04 | -5.4547E-04 | -1.9212E-06 |

In the 7th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 7th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 13 and Table 14 as the following values and satisfy the following conditions:

| 7th Embodiment |      |                       |       |
|----------------|------|-----------------------|-------|
| f [mm]         | 3.43 | (R1 + R2)/(R1 - R2)   | -1.36 |
| Fno            | 2.22 | (R3 + R4)/(R3 - R4)   | 0.65  |
| HFOV [deg.]    | 40.0 | (R9 - R10)/(R9 + R10) | 0.96  |
| V3/V2          | 0.36 | f/R1 + f/R2           | 1.55  |

-continued

| 7th Embodiment |       |       |      |
|----------------|-------|-------|------|
| (V3 + V6)/V5   | 1.36  | f2/f3 | 1.00 |
| CT2/T12        | 16.87 | f3/f4 | 0.05 |
| Td/ΣAT         | 3.92  | f1/f2 | 1.08 |
| ImgH/f         | 0.86  |       |      |

Furthermore, according to the 7th embodiment, when the focal length of the first lens element **710** is f1, the focal

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length of the second lens element **720** is  $f_2$ , the focal length of the third lens element **730** is  $f_3$ , the focal length of the fourth lens element **740** is  $f_4$ , a focal length of the fifth lens element **750** is  $f_5$ , and a focal length of the sixth lens element **760** is  $f_6$ , the following conditions are satisfied:  $|f_1| < |f_4|$ ;  $|f_2| < |f_4|$ ;  $|f_3| < |f_4|$ ;  $|f_5| < |f_4|$ ; and  $|f_6| < |f_4|$ .

#### 8th Embodiment

FIG. **15** is a schematic view of an image capturing device according to the 8th embodiment of the present disclosure. FIG. **16** shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 8th embodiment. In FIG. **15**, the image capturing device includes an optical image lens assembly (its reference numeral is omitted) and an image sensor **890**. The optical image lens assembly includes, in order from an object side to an image side, an aperture stop **800**, a first lens element **810**, a second lens element **820**, a third lens element **830**, a fourth lens element **840**, a fifth lens element **850**, a sixth lens element **860**, an IR-cut filter **870** and an image surface **880**, wherein the image sensor **890** is disposed on the image surface **880** of the optical image lens assembly. The optical image lens assembly has a total of six lens elements (**810-860**), and there is an air space between every two lens elements of the first lens element **810**, the second lens element **820**, the third lens element **830**, the fourth lens element **840**, the fifth lens element **850** and the sixth lens element **860** that are adjacent to each other.

The first lens element **810** with positive refractive power has an object-side surface **811** being convex in a paraxial region thereof and an image-side surface **812** being convex in a paraxial region thereof. The first lens element **810** is made of a plastic material, and has the object-side surface **811** and the image-side surface **812** being both aspheric.

The second lens element **820** with positive refractive power has an object-side surface **821** being concave in a paraxial region thereof and an image-side surface **822** being convex in a paraxial region thereof. The second lens element

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**820** is made of a plastic material, and has the object-side surface **821** and the image-side surface **822** being both aspheric.

The third lens element **830** with negative refractive power has an object-side surface **831** being convex in a paraxial region thereof and an image-side surface **832** being concave in a paraxial region thereof. The third lens element **830** is made of a plastic material, and has the object-side surface **831** and the image-side surface **832** being both aspheric.

The fourth lens element **840** with positive refractive power has an object-side surface **841** being convex in a paraxial region thereof and an image-side surface **842** being concave in a paraxial region thereof. The fourth lens element **840** is made of a plastic material, and has the object-side surface **841** and the image-side surface **842** being both aspheric. Furthermore, the image-side surface **842** of the fourth lens element **840** includes at least one convex shape in an off-axial region thereof.

The fifth lens element **850** with positive refractive power has an object-side surface **851** being concave in a paraxial region thereof and an image-side surface **852** being convex in a paraxial region thereof. The fifth lens element **850** is made of a plastic material, and has the object-side surface **851** and the image-side surface **852** being both aspheric.

The sixth lens element **860** with negative refractive power has an object-side surface **861** being convex in a paraxial region thereof and an image-side surface **862** being concave in a paraxial region thereof. The sixth lens element **860** is made of a plastic material, and has the object-side surface **861** and the image-side surface **862** being both aspheric. Furthermore, the image-side surface **862** of the sixth lens element **860** includes at least one convex shape in an off-axial region thereof.

The IR-cut filter **870** is made of a glass material and located between the sixth lens element **860** and the image surface **880**, and will not affect the focal length of the optical image lens assembly.

The detailed optical data of the 8th embodiment are shown in Table 15 and the aspheric surface data are shown in Table 16 below.

TABLE 15

| 8th Embodiment                            |               |                  |           |          |       |        |              |  |
|-------------------------------------------|---------------|------------------|-----------|----------|-------|--------|--------------|--|
| f = 3.29 mm, Fno = 2.22, HFOV = 41.5 deg. |               |                  |           |          |       |        |              |  |
| Surface #                                 |               | Curvature Radius | Thickness | Material | Index | Abbe # | Focal Length |  |
| 0                                         | Object        | Plano            | Infinity  |          |       |        |              |  |
| 1                                         | Ape. Stop     | Plano            | -0.043    |          |       |        |              |  |
| 2                                         | Lens 1        | 2.960 ASP        | 0.400     | Plastic  | 1.544 | 55.9   | 4.40         |  |
| 3                                         |               | -11.862 ASP      | 0.030     |          |       |        |              |  |
| 4                                         | Lens 2        | -32.493 ASP      | 0.516     | Plastic  | 1.544 | 55.9   | 6.50         |  |
| 5                                         |               | -3.208 ASP       | 0.050     |          |       |        |              |  |
| 6                                         | Lens 3        | 3.967 ASP        | 0.240     | Plastic  | 1.660 | 20.4   | -5.10        |  |
| 7                                         |               | 1.778 ASP        | 0.397     |          |       |        |              |  |
| 8                                         | Lens 4        | 5.046 ASP        | 0.264     | Plastic  | 1.639 | 23.5   | 138.77       |  |
| 9                                         |               | 5.242 ASP        | 0.332     |          |       |        |              |  |
| 10                                        | Lens 5        | -76.201 ASP      | 0.652     | Plastic  | 1.544 | 55.9   | 2.48         |  |
| 11                                        |               | -1.328 ASP       | 0.351     |          |       |        |              |  |
| 12                                        | Lens 6        | 2.529 ASP        | 0.320     | Plastic  | 1.544 | 55.9   | -1.90        |  |
| 13                                        |               | 0.702 ASP        | 0.400     |          |       |        |              |  |
| 14                                        | IR-cut filter | Plano            | 0.210     | Glass    | 1.517 | 64.2   | —            |  |
| 15                                        |               | Plano            | 0.254     |          |       |        |              |  |
| 16                                        | Image         | Plano            | —         |          |       |        |              |  |

Reference wavelength is 587.6 nm (d-line).

TABLE 16

| Aspheric Coefficients |             |             |             |             |             |             |
|-----------------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Surface #             | 2           | 3           | 4           | 5           | 6           | 7           |
| k =                   | -1.1191E+01 | -5.1448E+00 | -4.0466E+01 | -6.4952E+01 | -1.9137E+01 | 2.4372E-01  |
| A4 =                  | -5.5418E-02 | -2.1487E-01 | -7.5396E-02 | -2.2352E-01 | -1.5720E-01 | -2.1598E-01 |
| A6 =                  | -3.9164E-02 | 3.7182E-01  | 3.8424E-01  | 5.3300E-01  | 1.9750E-01  | 2.4977E-01  |
| A8 =                  | -3.2376E-01 | -2.3442E-01 | -2.5733E-02 | -1.1428E+00 | -4.8363E-01 | -3.8034E-01 |
| A10 =                 | 9.3809E-01  | 8.7399E-02  | -4.3303E-01 | 1.6575E+00  | 8.9133E-01  | 4.7185E-01  |
| A12 =                 | -1.1734E+00 | -1.6323E-01 | 3.2092E-01  | -1.3004E+00 | -7.8094E-01 | -3.1876E-01 |
| A14 =                 | 6.1527E-01  | 1.2622E-01  | -6.4058E-02 | 3.9534E-01  | 2.4965E-01  | 8.5993E-02  |
| Surface #             | 8           | 9           | 10          | 11          | 12          | 13          |
| k =                   | 1.9977E-01  | 5.5806E+00  | -1.0001E+00 | -1.5413E+00 | -9.0000E+01 | -5.1191E+00 |
| A4 =                  | -5.6465E-02 | 1.9136E-02  | 2.7675E-01  | 3.5799E-01  | -1.8875E-01 | -1.0147E-01 |
| A6 =                  | -2.5889E-01 | -4.5395E-01 | -3.5814E-01 | -2.6890E-01 | 9.9525E-02  | 4.3791E-02  |
| A8 =                  | 6.2364E-01  | 6.8827E-01  | 2.2588E-01  | 8.2765E-02  | -1.3837E-01 | -1.6897E-02 |
| A10 =                 | -6.7568E-01 | -5.3274E-01 | -7.8624E-02 | -1.3616E-03 | 9.5851E-02  | 4.2777E-03  |
| A12 =                 | 3.5560E-01  | 2.0703E-01  | 1.0388E-02  | -5.0157E-03 | -2.9948E-02 | -6.1586E-04 |
| A14 =                 | -7.3054E-02 | -3.0898E-02 | 1.0372E-03  | 1.0902E-03  | 4.4107E-03  | 4.2780E-05  |
| A16 =                 |             | 3.5438E-04  | -2.8753E-04 | -7.3801E-05 | -2.5220E-04 | -1.0073E-06 |

In the 8th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 8th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 15 and Table 16 as the following values and satisfy the following conditions:

| 8th Embodiment  |       |                         |       |
|-----------------|-------|-------------------------|-------|
| f [mm]          | 3.29  | $(R1 + R2)/(R1 - R2)$   | -0.60 |
| Fno             | 2.22  | $(R3 + R4)/(R3 - R4)$   | 1.22  |
| HFOV [deg.]     | 41.5  | $(R9 - R10)/(R9 + R10)$ | 0.97  |
| V3/V2           | 0.36  | $f/R1 + f/R2$           | 1.39  |
| (V3 + V6)/V5    | 1.36  | $ f2/f3 $               | 1.27  |
| CT2/T12         | 17.20 | $ f3/f4 $               | 0.04  |
| Td/ $\Sigma$ AT | 3.06  | $f1/f2$                 | 0.68  |
| ImgH/f          | 0.89  |                         |       |

Furthermore, according to the 8th embodiment, when the focal length of the first lens element **810** is  $f1$ , the focal length of the second lens element **820** is  $f2$ , the focal length of the third lens element **830** is  $f3$ , the focal length of the fourth lens element **840** is  $f4$ , a focal length of the fifth lens element **850** is  $f5$ , and a focal length of the sixth lens element **860** is  $f6$ , the following conditions are satisfied:  $|f1| < |f4|$ ;  $|f2| < |f4|$ ;  $|f3| < |f4|$ ;  $|f5| < |f4|$ ; and  $|f6| < |f4|$ .

#### 9th Embodiment

FIG. **17** is a schematic view of an electronic device **10** according to the 9th embodiment of the present disclosure. The electronic device **10** of the 9th embodiment is a smart-phone, wherein the electronic device **10** includes an image capturing device **11**. The image capturing device **11** includes an optical image lens assembly (its reference numeral is omitted) according to the present disclosure and an image sensor (its reference numeral is omitted), wherein the image sensor is disposed on an image surface of the optical image lens assembly.

#### 10th Embodiment

FIG. **18** is a schematic view of an electronic device **20** according to the 10th embodiment of the present disclosure. The electronic device **20** of the 10th embodiment is a tablet personal computer, wherein the electronic device **20** includes an image capturing device **21**. The image capturing device **21** includes an optical image lens assembly (its reference numeral is omitted) according to the present disclosure and an image sensor (its reference numeral is omitted), wherein the image sensor is disposed on an image surface of the optical image lens assembly.

#### 11th Embodiment

FIG. **19** is a schematic view of an electronic device **30** according to the 11th embodiment of the present disclosure. The electronic device **30** of the 11th embodiment is a wearable device, wherein the electronic device **30** includes an image capturing device **31**. The image capturing device **31** includes an optical image lens assembly (its reference numeral is omitted) according to the present disclosure and an image sensor (its reference numeral is omitted), wherein the image sensor is disposed on an image surface of the optical image lens assembly.

The foregoing description, for purpose of explanation, has been described with reference to specific embodiments. It is to be noted that Tables 1-16 show different data of the different embodiments; however, the data of the different embodiments are obtained from experiments. The embodiments were chosen and described in order to best explain the principles of the disclosure and its practical applications, to thereby enable others skilled in the art to best utilize the disclosure and various embodiments with various modifications as are suited to the particular use contemplated. The embodiments depicted above and the appended drawings are exemplary and are not intended to be exhaustive or to limit the scope of the present disclosure to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings.

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What is claimed is:

1. An optical image lens assembly comprising, in order from an object side to an image side:

a first lens element with positive refractive power having an object-side surface being convex in a paraxial region thereof;

a second lens element with positive refractive power having an image-side surface being convex in a paraxial region thereof;

a third lens element;

a fourth lens element having an image-side surface being concave in a paraxial region thereof;

a fifth lens element with positive refractive power having an object-side surface being concave in a paraxial region thereof and an image-side surface being convex in a paraxial region thereof, wherein the object-side surface and the image-side surface of the fifth lens element are aspheric; and

a sixth lens element with negative refractive power having an image-side surface being concave in a paraxial region thereof, wherein an object-side surface and the image-side surface of the sixth lens element are aspheric, and the image-side surface of the sixth lens element comprises at least one convex shape in an off-axial region thereof;

wherein the optical image lens assembly has a total of six lens elements, there is an air space between every two lens elements of the first lens element, the second lens element, the third lens element, the fourth lens element, the fifth lens element and the sixth lens element that are adjacent to each other, a curvature radius of an object-side surface of the second lens element is R3, a curvature radius of the image-side surface of the second lens element is R4, a focal length of the second lens element is f2, a focal length of the third lens element is f3, a focal length of the fourth lens element is f4, an axial distance between the object-side surface of the first lens element and the image-side surface of the sixth lens element is Td, a sum of axial distances between every two of the lens elements of the optical image lens assembly that are adjacent to each other is  $\Sigma AT$ , and the following conditions are satisfied:

$$0.53 \leq (R3+R4)/(R3-R4);$$

$$|f2/f3| < 3.0;$$

$$|f3/f4| < 10; \text{ and}$$

$$4.48 \leq Td/\Sigma AT < 20.$$

2. The optical image lens assembly of claim 1, wherein the third lens element with negative refractive power has an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof, and the object-side surface and the image-side surface of the third lens element are aspheric.

3. The optical image lens assembly of claim 1, wherein the first lens element has an image-side surface being concave in a paraxial region thereof, a curvature radius of the object-side surface of the first lens element is R1, a curvature radius of the image-side surface of the first lens element is R2, and the following condition is satisfied:

$$(R1+R2)/(R1-R2) < -1.60.$$

4. The optical image lens assembly of claim 1, wherein the following condition is satisfied:

$$0 < |f2/f3| < 1.50.$$

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5. The optical image lens assembly of claim 1, wherein the image-side surface of the fourth lens element comprises at least one convex shape in an off-axial region thereof.

6. The optical image lens assembly of claim 1, wherein the third lens element has an image-side surface comprising at least one convex shape in an off-axial region thereof.

7. The optical image lens assembly of claim 1, wherein a focal length of the first lens element is f1, and the following condition is satisfied:

$$0.50 < f1/f2 < 2.0.$$

8. The optical image lens assembly of claim 1, wherein a curvature radius of the object-side surface of the fifth lens element is R9, a curvature radius of the image-side surface of the fifth lens element is R10, and the following condition is satisfied:

$$0 < (R9-R10)/(R9+R10) < 0.75.$$

9. The optical image lens assembly of claim 1, wherein a focal length of the first lens element is f1, a focal length of the fifth lens element is f5, a focal length of the sixth lens element is f6, and the following conditions are satisfied:

$$|f1| < |f4|;$$

$$|f2| < |f4|;$$

$$|f3| < |f4|;$$

$$|f5| < |f4|; \text{ and}$$

$$|f6| < |f4|.$$

10. The optical image lens assembly of claim 1, wherein the following condition is satisfied:

$$|f3/f4| < 2.0.$$

11. The optical image lens assembly of claim 1, wherein the following condition is satisfied:

$$0.80 < (R3+R4)/(R3-R4) < 4.0.$$

12. The optical image lens assembly of claim 1, wherein a central thickness of the second lens element is CT2, an axial distance between the first lens element and the second lens element is T12, and the following condition is satisfied:

$$0 < CT2/T12 < 5.0.$$

13. The optical image lens assembly of claim 1, wherein an Abbe number of the second lens element is V2, an Abbe number of the third lens element is V3, and the following condition is satisfied:

$$0.20 < V3/V2 < 0.60.$$

14. The optical image lens assembly of claim 1, wherein a focal length of the optical image lens assembly is f, a curvature radius of the object-side surface of the first lens element is R1, a curvature radius of an image-side surface of the first lens element is R2, and the following condition is satisfied:

$$1.0 < f/R1 + f/1R2 < 5.0.$$

15. An image capturing device, comprising:  
the optical image lens assembly of claim 1; and  
an image sensor, wherein the image sensor is disposed on an image surface of the optical image lens assembly.

16. An electronic device, comprising:  
the image capturing device of claim 15.

\* \* \* \* \*